Adaptive Visual Noise Cancellation for Attention Guidance on a Consumer Passthrough Headset

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Supervisor: John Dingliana

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

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August 6, 2026

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Extended-reality headsets are conventionally used to *add* information to the world. This dissertation investigates the opposite operation: using a consumer video-see-through (VST) headset to *subtract*, dimming, muting, or re-grading the visual periphery so that attention is guided towards, and held on, a user-chosen region of the real world. The work asks one question with two halves. Can subtractive attention guidance be delivered on consumer XR hardware, and, when it is delivered, does it actually help human attention?

The technical half confronts a defining constraint of the Meta Quest 3 platform: the operating system owns the passthrough layer, which applications may neither read nor spatially modify. The dissertation contributes a three-tier design space of diminished reality (DR) under this compositing constraint, and a working reference implementation spanning it. A world-locked *focus window* is rendered as an absence in an overlay sphere, so native passthrough passes through it at full quality, while the periphery carries the manipulation. Two peripheral treatments are carried to confirmatory evaluation: Sign-Pop, which re-grades salience gated on detected objects, and Hard Dark, which occludes the periphery completely. A structured prior-art search found no published or shipped precedent for the system's hybrid composite of native passthrough and a shader-processed camera feed. A technical evaluation plan covering frame cost, latency, legibility across rendering paths, and the gap between live and oracle object detection is specified, with preliminary results from the offline detection pipeline.

The human half contributes a pre-registered, fully within-subjects user study, built to be falsifiable rather than merely hopeful: a workstation distractor-suppression paradigm carrying the single primary hypothesis, a driving-scene marker-detection paradigm with an equivalence-bounded situational-awareness safety test, and a pre-committed decision table under which every outcome is a conclusive result. Recruitment is ongoing against a target

of twenty. Fifteen people have been tested, giving fourteen usable paired participants in the workstation block and thirteen in the driving block. Interim results show a 3.34-point accuracy gain on the workstation task (95% CI +0.33 to +6.34, p = .032), no loss of peripheral awareness, and a detection benefit of 16.5 points concentrated in the 20–30° eccentricity band that the system's own detection-gated geometry predicts. One methodological wrinkle is reported rather than smoothed over. The safety hypothesis was pre-registered as a two-sided equivalence test, which is not established at this sample size because a benefit larger than the margin cannot be excluded, even though the one-directional safety question it was written to answer is answered cleanly. All results are reported and discussed as interim findings, not as the final confirmatory outcome.

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Chapter 1

Introduction

1.1 Motivation: attention is the scarce resource

In the Mahābhārata, Droṇa, the martial teacher of the Pāṇḍava and Kaurava princes, sets up a wooden bird in a tree and tells his students to aim for its eye. Before each releases the arrow, he asks what they see. One says the tree. Another the branches, the leaves, his brothers standing beside him. Droṇa asks Arjuna the same question, then asks again: does he see the tree? The branches? Arjuna answers no to each, until only one thing remains: "I see only the eye of the bird." For that moment, the tree, the branches, and everyone else standing there have not merely fallen out of Arjuna's interest. They have gone. This dissertation pursues the same operation on a headset instead of a bow: not adding emphasis to what matters, but removing everything that does not, until it is no longer competing to be seen.

The modern visual environment is engineered against sustained attention. Open-plan offices place moving colleagues at the edge of vision. Shared homes put televisions and phone screens beside workspaces. Streetscapes compete for a driver's gaze with advertising designed by professionals to win it. The cost is well documented: task-irrelevant peripheral events capture attention involuntarily, and each capture carries a performance penalty on the task the viewer intended to do. In immersive settings the effect is directly measurable, introducing peripheral visual distractors into a virtual classroom more than doubled commission errors on a sustained-attention task in a recent study of sixty-six participants (Ai et al., 2025), replicating decades of laboratory findings on involuntary attentional capture by peripheral motion and luminance change (Itti et al., 1998).

The conventional response of the extended-reality (XR) community to attention problems has been *additive*: arrows, halos, attention funnels, and highlights layered onto the world to point at what matters (Biocca et al., 2006). This dissertation pursues the complementary and far less explored operation: *subtraction*. If the periphery pulls attention away, dim it, mute it, or re-grade it so that it pulls less. In the literature this family of operations is called diminished reality (DR), concealing or attenuating real-world content rather than adding to it (Mori et al., 2017; Cheng et al., 2022), and its application to distraction has recently been named *visual noise cancellation*, by direct analogy with the acoustic kind (Hong et al., 2024).

A video-see-through (VST) headset is, in principle, the ideal instrument for visual noise cancellation. Unlike optical see-through (OST) glasses, whose additive light engines physically cannot darken the world, a VST headset re-displays the world on an opaque screen: every pixel the user sees is, in principle, available for modification. The Meta Quest 3, an affordable, widely owned consumer device with colour passthrough, would seem to make DR-based attention guidance deployable at scale for the first time. In principle.

1.2 The compositing constraint

In practice, the premise "every pixel is available for modification" is false on consumer hardware, and that falsity is the motivating technical problem of this dissertation.

On the Quest 3, the passthrough view of the world is composited by the operating system, not by the application. An application cannot read the passthrough layer, cannot shade it, and cannot spatially mask it. It may only draw *on top* of it, or apply a global colour transform through a narrow styling interface. Since early 2025 the platform has additionally exposed the raw forward camera feed to applications (Passthrough Camera Access), but at materially lower quality than the system passthrough the user normally sees: a monocular 1280×960 stream covering roughly 85–90° of field of view, versus the system layer's full coverage of the headset's approximately 110° display, with its correct per-eye reprojection. The one place where full pixel control exists is therefore also the place where image quality is worst. A recent psychophysical study shows the stakes plainly: no current VST headset reaches normal human visual acuity through its passthrough, with the Quest 3 degrading further in low light (Wang et al., 2026).

Subtractive attention guidance on this platform is consequently a *compositing-rights problem*. The effect one wants: "dim everything except the region that matters, without degrading the region that matters", cannot be obtained by editing the image the user sees, because no application is permitted to edit it. Chapter 3 shows that what remains is a narrow but genuinely usable design space, organised into three tiers of access, and Chapter 4 presents a working system that exploits an architectural inversion at its centre: the region that matters is rendered as an *absence*, a transparent window in an overlay, so that the highest-quality image on the device, the native system passthrough, serves the focus region for free, while the manipulation is confined to the periphery where acuity is lower and precision matters less. On the same principle, the system's most capable mode composites native passthrough inside the window with a shader-processed camera feed outside it, a hybrid for which a structured prior-art search (Chapter 2) found no published or shipped precedent.

1.3 One question, two halves

The dissertation asks a single question, with two distinct halves:

> **Can subtractive attention guidance be delivered on consumer XR hardware? And when it is delivered, does it actually help human attention?**

The two halves of the question structure the entire work.

Part T (technical): can it be built? What degree of diminishment control is achievable on a consumer VST headset without OS-level access to the passthrough layer, and at what cost to image quality, field of view, latency, and comfort? Part T is answered by the design space (Chapter 3), the reference implementation (Chapter 4), and the technical evaluation plan with preliminary results (Chapter 5).

Part H (human): does it work? Under controlled distraction, does peripheral diminishment reduce the cost distractors impose on a focal task, and does salience regrading enhance focal detection without unacceptably degrading peripheral awareness? Part H is answered by a pre-registered user study design (Chapter 6).

Two methodological commitments connect the halves and should be named at the outset, because they are choices rather than accidents.

First, the study's dynamic scenario runs as a *simulation*, pre-recorded driving footage presented on the system's rendering sphere rather than live passthrough of a street, following the closest published DR evaluations, which simulated diminishment inside virtual environments because controlled, repeatable, event-dense stimuli cannot be obtained from the uninstrumented real world (Cheng et al., 2022; McLaughlin et al., 2025). Second, detection is supplied by an *oracle* computed offline over the study footage rather than by a model running on the headset, so the study can ask whether an effect built on excellent detection helps attention without first engineering excellent detection on-device. Chapter 5 measures how far a live pipeline currently falls short of that oracle, and Chapter 6 carries both commitments forward as stated limitations.

The two study blocks bracket the feasibility spectrum. The primary, workstation, block runs on *real passthrough with real physical distractors*, the actual artefact with no simulation. The secondary, driving, block evaluates the concept in the simulated deployment context.

1.4 Research questions and hypotheses

Three research questions operationalise the spine question. They are stated here in summary. Their full pre-registered form, including smallest effect sizes of interest, equivalence margins, statistical tests, and a decision table committing the dissertation to a conclusion under every outcome, is given in Chapter 6.

- **RQ1 (primary).** Does peripheral DR dimming (Hard Dark, world-locked window) reduce the performance cost that peripheral visual distractors impose on a focal workstation task? *Hypothesis H1:* the error rate on the workstation task, run with distractors present throughout, is lower with the vignette on than off, by at least a pre-registered fraction of the vignette-off error rate.

- **RQ2 (secondary).** Does SignPop's detection-gated salience re-grading improve detection specifically where the gate has room to act, without degrading detection pooled across the whole scene beyond an acceptable margin? *Hypotheses H2a/H2b:* detection improves in the eccentricity band where the colour-pop gate is strong but the target is still within useful glance range (H2a), and hit rate pooled across the whole field of view is equivalent to the vignette-off condition within a 10-percentage-point margin (H2b). H2b is a safety hypothesis tested by equivalence, so failing it would be a conclusive negative finding rather than an ambiguous null.

- **RQ3 (exploratory).** After experiencing the system, what do participants report about perceived focus benefit, perceived awareness cost, and visual comfort?

The design philosophy behind these hypotheses deserves one sentence here: the study is built so that it cannot return "inconclusive". Distractors are already known to impose a measurable cost, on the evidence of the sixty-six-participant result cited above. The study takes that cost as given, asks whether the system reduces it, and pre-commits to the interpretation of every outcome, including failure.

1.5 Contributions

- **C1: Design space.** A taxonomy of DR capability under compositing constraints on consumer VST hardware: three access tiers, with the achievable effects, ceilings, and costs of each, characterised empirically rather than speculatively (Chapter 3).
- **C2: Reference implementation.** An open, working multi-mode DR attention-guidance system on Quest 3 passthrough: the world-locked focus-window-as-absence architecture, world-direction fusion sampling that lets a monocular camera feed fuse binocularly, motion-coupled comfort suppression, and ten peripheral treatments spanning the three-tier space. Two of them, SignPop (detection-gated salience regrading) and Hard Dark (full peripheral occlusion), are carried to confirmatory evaluation, with documented failure modes (Chapter 4).
- **C3: Technical evaluation.** A benchmark plan (frame cost, latency, legibility across rendering paths, thermal endurance, and the oracle-versus-live detection gap) with preliminary results from the offline detection pipeline (Chapter 5).
- **C4: Pre-registered user study, with interim results.** A falsifiable within-subjects design ($N = 20$) with a single properly powered primary hypothesis, equivalence-bounded safety testing, pilot gates, and a pre-committed decision table, together with an interim analysis of the fourteen participant pairs collected so far: a workstation-focus benefit (+3.34 points accuracy), no loss of peripheral awareness, and a detection gain specifically in the eccentricity band the system's own design predicts (Chapter 6).

1.6 Scope: what this dissertation does not claim

Because the credibility of a small, carefully scoped study depends on the discipline of its claims, the non-claims are stated as prominently as the claims.

No driving or road-safety claims. The driving footage used in the secondary study block is a dynamic, high-event-rate monitoring *stimulus*. A seated participant passively watching passenger-perspective video licenses no inference about driving, drivers, or road safety. Such inference would require a driving simulator with vehicle control, which is out of scope.

No product claims. The study evaluates attention mechanisms under controlled distraction, not the desirability or readiness of a wearable product. Comfort findings from a 2026 headset transfer to future lighter hardware only as bounds, and are reported as such.

Oracle conditioning stays visible. SignPop's finding depends on detection: on offline, oracle-quality detection, not on live, on-device detection. Every place this dissertation reasons about a live or on-device detector, that reasoning is conditioned on the oracle bound that Chapter 5 measures, and marked accordingly. Oracle performance is never substituted for live performance without saying so.

Interim, not final, results. At the time of writing the technical benchmark campaign of Chapter 5 has not been run, beyond the offline detection bake reported there as a preliminary result. The user study of Chapter 6 is a different matter. It is actively collecting, against the pre-registered design that chapter specifies, and eleven usable pairs exist for each of its two confirmatory blocks against a target of twenty. Chapter 6, Section 6.11 reports what those eleven pairs show, explicitly as an interim analysis. Recruitment continues, achieved power at this N is below the pre-registered target, and every claim built on it in this dissertation is qualified as interim accordingly. No result in this document is asserted as the pre-registered confirmatory finding. Where an interim result is discussed, it is named as interim every time.

1.7 Roadmap

Chapter 2 surveys the literature across diminished reality, saliency modulation and gaze guidance, foveated rendering, attention guidance in driving and workstation contexts, and human-factors foundations, closing with the research gaps this work addresses and a documented prior-art search for the system's core architecture. Chapter 3 develops the three-tier design space of DR under compositing constraints. Chapter 4 presents the system: architecture, the full mode family and the three modes the study uses, interaction design, and the engineering record including dead ends. Chapter 5 specifies the technical evaluation and reports preliminary detection-pipeline results. Chapter 6 presents the pre-registered user study design in full and reports interim results from the fourteen Block A and thirteen Block B pairs collected so far. Chapter 7 discusses these interim results, threats to validity, and transfer, including what does and does not carry to optical see-through hardware. Chapter 8 concludes.

The three peripheral treatments seen from the user's viewpoint: SignPop, Soft Dark, Hard Dark, each with the world-locked focus window visible. Composite of headset screenshots.

Figure 1.1: The three peripheral treatments seen from the user's viewpoint.

The two-part structure of the dissertation: Part T (design space → system → benchmarks) and Part H (pre-registered study), with the workstation block bridging them on real hardware.

Figure 1.2: The two-part structure of the dissertation.

Chapter 2

Background and Related Work

This chapter argues one line of reasoning end to end, across eight sections, rather than surveying eight bodies of literature side by side. In order: is the problem real, what has already been tried, has removing things instead of adding them been tried and did it work, can any of this be built on hardware people actually own, what does a system built that way need to be tested for, and how should that test be run.

Section 2.1 establishes why peripheral events capture attention involuntarily and why that capture costs something. Section 2.2 reviews the techniques already used to guide attention in mediated views, and what they were found to cost. Section 2.3 asks whether removing things rather than adding them has been tried, covering diminished reality (DR) and visual noise cancellation, the conceptual home of this work. Section 2.4 asks what a human periphery tolerates and where the comfort limits sit. Section 2.5 asks what consumer passthrough hardware actually permits, and reports a documented search establishing that this dissertation's architecture has no direct precedent. Sections 2.6 and 2.7 cover the applied contexts that supply the test scenarios and the methods used to evaluate them. Section 2.8 draws out the research gaps and positions this work against its two nearest published neighbours.

Throughout, the chapter keeps a specific discipline. A study is reported here when it establishes that the problem is real, when it justifies a value this system actually uses, or when it supplies a method the evaluation adopts. Work that merely neighbours the topic is left to the surveys that cover it.

2.1 Foundations: Attention, Salience, and Cognitive Load

2.1.1 Salience and attentional capture

The computational account of visual attention that underpins this entire dissertation is the salience-map model of Itti et al. (1998): attention is drawn to locations whose centre–surround contrast, in intensity, colour opponency, or orientation, differs most from their neighbourhood. Three consequences of this model recur throughout the design of the present system. First, *contrast is the master variable*: any manipulation that raises local contrast attracts gaze, and any manipulation that flattens it repels gaze. Second, salience is channel-specific, and the channels are not equal. In every head-to-head comparison in the guidance literature, luminance manipulations outperform chromatic ones (Bailey et al., 2009; Grogorick et al., 2017). This is why every mode in this system dims before it does anything else, and why the one mode that relies on colour alone is not the one carried to confirmatory testing. Third, salience operates *pre-attentively*: capture by a high-contrast peripheral event happens before, and regardless of, the observer's intentions. A periphery whose salience has been flattened stops issuing capture events at all, which is the mechanism the dark modes exploit.

Duan et al. (2022) extend the account to augmented reality and show that the opacity of AR content directly and predictably modulates where attention lands. Opacity is therefore a dose, not a style, which is what makes the attenuation axis of Chapter 3 a design axis with a middle rather than an on/off switch.

2.1.2 Cognitive load

Where salience explains *how* the periphery captures attention, Cognitive Load Theory explains *why that capture is costly*. Extraneous load, processing demanded by information irrelevant to the task, competes for the same limited working-memory resources as the task itself. Buchner et al. (2022) review the evidence that AR can reduce, not only add, cognitive load when it is used to structure or filter what the user must process. This supplies the theoretical justification for the entire project: peripheral diminishment is an *extraneous-load intervention*. It does not make the user smarter or the task easier. It removes competing processing demands at the perceptual source. The dissertation's primary hypothesis (H1, Chapter 6) is this theory operationalised: if peripheral distractors impose a measurable performance cost on a focal task, a manipulation that suppresses their salience should reduce that cost.

2.1.3 The inherent tension

This dissertation pairs two hypotheses, not one, because of a tension the literature above already names. H1 says suppression reduces distraction cost. It says nothing about what suppression might cost in return, and the same literature that motivates diminishment also names its central risk. Attention capture by peripheral events is not a design flaw of the human visual system. It is a safety mechanism. Suppressing it trades focus against situational awareness, a tension documented empirically in the DR literature (Murph et al., 2021; McLaughlin et al., 2025) and formalised in this dissertation as an equivalence-tested safety hypothesis rather than an afterthought (H2b, Chapter 6). This tension, which every section of this chapter returns to, is the reason the dissertation's evaluation design refuses to measure benefit without simultaneously measuring cost.

2.2 Saliency Modulation and Gaze Guidance in Mediated Views

The techniques for steering attention in a mediated view form a spectrum from imperceptible to overt. This section traverses it: subtle gaze direction (2.2.1), saliency modulation of video see-through imagery, the direct ancestor of ColorPop (2.2.2), and overt cues with the comparative studies that justify combining techniques (2.2.3).

2.2.1 Subtle gaze direction

The subtle gaze direction (SGD) family originates with Bailey et al. (2009): a brief luminance modulation ($\pm 9.5\%$, $\sim 0.76°$ region, 10 Hz) presented in the viewer's periphery attracts a saccade, and the cue is terminated the moment gaze moves toward it, so the viewer never foveates the modulation and remains unaware of being steered. Luminance modulation outperformed warm–cool chromatic modulation, an early instance of the luminance-dominance law of Section 2.1. McNamara et al. (2008) demonstrated task-level value: in visual search, subtle modulation raised accuracy from 40.97% to 56.25%, statistically indistinguishable from an *obvious* cue (56.94%), with zero of eighteen participants noticing the manipulation. The paradigm has since been extended to clinical training, to automatic target prediction, and to stereo depth, but always as an *added* transient cue.

Two results from this family set hard constraints on the present design. Grogorick et al. (2018) ran the most complete comparison of the subtlety–effectiveness terrain, five guidance methods in an immersive dome with $N = 102$. Local magnification achieved the best guidance-per-noticeability ratio, while *SpatialBlur*, whole-periphery blur, was the least subtle and actively irritated 59 of 102 participants. Their conclusion, "no method remains completely unnoticed", together with the blur-irritation finding, decided two things here. Whole-periphery blur is not among the modes carried forward (Chapter 4), and the design accepts noticeability rather than chasing imperceptibility. Second, Erickson et al. (2023)'s parametric analysis of hue versus saturation versus value found hue shifts the most confusing of the three, which is why every mode in this system manipulates saturation and luminance and none of them shifts hue.

Slow ramps constitute the second escape hatch from the noticeability trade-off: Hata et al. (2016) showed that blur introduced gradually enough stays below awareness threshold while still biasing gaze, because the guidance threshold sits below the awareness threshold. This finding is the perceptual justification for this system's formation animation: effects ramp in over seconds rather than switching on (Chapter 4).

2.2.2 Saliency modulation of see-through video

Where SGD adds a transient stimulus, saliency *modulation* re-grades the image itself, raising conspicuity where attention should go and lowering it elsewhere. This is the direct technical lineage of ColorPop.

The foundational demonstration is Veas et al. (2011): using an Itti-style conspicuity model, they raised salience in a focus region and suppressed it in context regions of video imagery (channel order: luminance first, then red–green, then blue–yellow; algorithm in Mendez et al., 2010). Modulated video was statistically indistinguishable from unmodulated video for naive viewers, yet produced faster first fixations on target regions ($t(35) = 2.916$, $p < .01$) and improved recall of modulated content ($0.19 \rightarrow 0.31$, $p = .008$). This remains the strongest evidence that video-mediated reality can be re-graded imperceptibly with measurable attentional consequences. And video mediation is exactly what a passthrough headset is. Kalkofen et al. (2007) contribute the focus-and-context vocabulary this dissertation uses throughout: the design discipline of keeping a focus region maximally legible while context is compressed but not destroyed.

The modern anchor is Sutton et al. (2022): saliency modulation of the *real world* viewed through AR glasses, implemented as saturation adjustment plus sigmoidal contrast remapping ($\alpha = 10$, $\beta = 0.5$), deliberately avoiding hue shifts, which participants found confusing. Modulation drew significantly more fixations to target regions ($\chi^2 = 25.57$, $p < .001$), but, an honest and important negative, was *not* imperceptible. Against an overt circle cue, modulation's advantage was that it preserved natural scene exploration where explicit markers "glue" gaze to themselves. This dissertation's ColorPop mode implements Sutton et al.'s exact sigmoidal contrast recipe on its salient-colour channel (Chapter 4), inheriting the philosophy: accept visibility, preserve exploration, re-grade rather than annotate. The critical difference is platform. Sutton et al. built a bench-mounted optical see-through rig whose additive optics can only add light. The present system runs on consumer video see-through hardware with full subtractive control of its rendered layers (Section 2.5).

An eye-tracked comparison in cinematic VR speaks to which *diminishment* channel works: comparing area darkening, context-based darkening, and desaturation for steering attention in 360° video, area darkening guided attention most effectively while desaturation alone had almost no guidance effect (Wang et al., 2020). It is a single result from one small-venue study, not a consensus finding, but it corroborates this project's mode hierarchy: the dark modes are the guidance workhorses, and desaturation appears only as one ingredient inside ColorPop's compound re-grading, never as a mode of its own.

2.2.3 Overt cues, comparisons, and combinations

At the overt end of the spectrum, the Attention Funnel (Biocca et al., 2006), a 3D tunnel of AR rings leading the head to an out-of-view target, improved search speed by 22% and consistency by 65% while reducing mental workload by 18%, establishing that aggressive guidance pays in speed what it costs in scene engagement. The corresponding failure mode is *attentional tunnelling*: excessive allocation of attention to guided content at the cost of neglecting the surrounding environment, which handheld-AR work has isolated experimentally as a function of task demand (Syiem et al., 2021). This is the risk most relevant to this dissertation's Hard Dark mode, and the reason its evaluation measures peripheral awareness rather than assuming it (H2b, Chapter 6). One further result shapes the architecture. Hein et al. (2019) report that combining a coarse guidance channel with a fine one significantly outperforms either alone, which is the pattern every mode here follows: coarse peripheral attenuation paired with a fine salience lift inside the focus window.

2.3 Diminished Reality and Visual Noise Cancellation

2.3.1 Definitions and taxonomy

Mori et al. (2017) provide the canonical survey and taxonomy of diminished reality: the family of techniques that *conceal, eliminate, or see through* real objects in a mediated view. Their survey establishes the field's legitimacy, that mediation of reality can subtract as well as add, and its historical centre of gravity: most classical DR work concerns object removal (inpainting a region so the object appears gone), which is computationally demanding and semantically targeted. The present work sits in a different corner of their taxonomy: it does not remove objects but *attenuates regions*, trading semantic precision for robustness, generality, and real-time feasibility on mobile hardware.

Cheng et al. (2022) supply the field's most direct study of how users *want* reality diminished. Across two studies (N = 16 and N = 12, conducted inside VR simulations of DR because no deployable DR hardware existed), they found participants preferred partial opacity reduction over complete removal, and wanted *context-preserving* diminishment: outlines or residual structure indicating that something is there, even when its detail is suppressed. Both findings are load-bearing for this dissertation: the preference for partial attenuation motivates the Soft Dark mode's deliberately incomplete dimming (maximum opacity ≈ 0.75, never a blackout), and the context-preservation finding motivates treating Hard Dark's total blackout as the *extreme* of the design space rather than its default. Equally load-bearing is Cheng et al.'s method: they could only simulate DR inside VR. This dissertation runs on the real passthrough hardware their participants could only imagine. The delta is developed in Section 2.8.

2.3.2 Visual noise cancellation

The most recent conceptual frame for this family of systems is *visual noise cancellation* (VNC): by analogy with acoustic noise cancellation, a head-worn display actively moderates the visual environment, attenuating "noise" while passing "signal" (Hong et al., 2024). The analogy is productive because it imports an architecture: acoustic ANC requires a microphone (sensing the environment), a processing stage, and a speaker (re-emitting a modified signal), precisely the camera → shader → display pipeline of a video-see-through headset. The VNC paper is the closest conceptual match to this entire project, and this dissertation can be read as the first *software-defined VNC system on consumer video-passthrough hardware*: each mode is a VNC profile, and the profiles differ in what they classify as noise and in how aggressively they attenuate it. The dark modes treat everything peripheral as noise. SignPop treats everything that is not a detected signal object as noise, so its rule is semantic rather than purely spatial. A follow-up line of work has begun evaluating VNC-style workspaces with sustained attention and working-memory batteries in mixed-reality settings (see Section 2.7).

2.3.3 DR for distraction suppression: the empirical record

The empirical case that diminishment helps focus is recent but consistent. McLaughlin et al. (2025) provide the closest published analogue to this dissertation's primary hypothesis: in two within-subject experiments (STEM graduates and NASA Johnson Space Center employees), participants performed a demanding assembly task under visual and auditory distraction, with DR-style attenuation of the distractors applied either universally or context-sensitively. Universal attenuation, the condition structurally closest to this project's peripheral vignette, produced fewer errors ($\beta = -0.37$, $p = .021$) and lower subjective workload than context-aware removal. Their analysis (linear mixed-effects models over trial-level data) is adopted directly by this dissertation's analysis plan. Critically, McLaughlin et al. also measured awareness of off-task objects and events, finding the focus/awareness trade-off real but manageable. This is the empirical ancestor of this work's H2b.

Lee and Kim (2025)'s DiminishAR approaches the same goal from the object side: AR-camouflaging a single distracting object (a smartphone) restored working-memory performance to a level statistically comparable with physically removing it ($N = 60$). Murph et al. document the same benefit–risk structure in a medical assembly context: DR lowered cognitive workload but risked situational awareness loss (Murph et al., 2021). In later work they describe methods of training users to overcome distraction under diminished reality, in which distractions are gradually reintroduced as the trainee's tolerance grows (Murph et al., 2022). That gradual-reintroduction logic is the source of this system's temporal fade transitions: effects ramp rather than switch, and they yield to head motion instead of fighting it (Chapter 4).

Two 2026 threads confirm both the demand for DR-for-attention and its immaturity. A co-design study with fifteen students with ADHD developed the concept of *attentional diminishment* through diary studies and workshops, with blur, desaturation, and removal as user-configurable filters, but built no working prototype (Exploring Diminished Reality for Attention Support, 2026). A CHI 2026 study on Apple Vision Pro obscured specific individuals to alleviate social discomfort, which is overlay-based diminishment on consumer hardware but targeted occlusion rather than peripheral attention guidance (Obscuring Undesirable Individuals, 2026). Neither implements peripheral diminishment for attention on real passthrough. Further back, FocalSpace (Yao et al., 2013) demonstrated the 2D ancestor of this project's workstation scenario: synthetic background blur in video conferencing, driven by depth and activity tracking, improving memory for meeting content and user preference.

2.4 Peripheral Vision, Foveated Degradation, and Comfort

2.4.1 What the periphery can lose

Peripheral vision is not blurry central vision. It is a differently tuned system with its own sensitivities, to motion and flicker above all, and its own tolerances. It notices contrast loss and motion change most, and chromatic detail loss least. Foveated rendering has mapped this terrain thoroughly, and its state-of-the-art survey (Wang et al., 2022) supplies the acceptable-degradation mathematics this project reuses for its focus-to-periphery falloffs. Tursun et al. (2019) showed that tolerable degradation is *content-dependent*, since a luminance-contrast-aware model permits far more degradation in low-contrast regions, which is the basis of the adaptive-intensity proposal in Chapter 8.

One perceptual finding is directly responsible for a parameter in this system. Patney et al. (2016) established that peripheral blur is detected primarily through the *contrast reduction* it causes, and that restoring local contrast after blurring roughly doubles the tolerable blur radius. This is implemented as `_BlurContrastRestore` (Chapter 4), and it also explains, mechanistically, Grogorick et al.'s irritation finding: naive whole-periphery blur announces itself through contrast collapse. That chain is why the dark modes attenuate luminance, which the periphery tolerates, rather than blurring structure, which it resents.

2.4.2 Comfort limits of peripheral restriction

A second literature approaches the periphery from the comfort side: field-of-view restriction as a cybersickness countermeasure. Fernandes and Feiner (2016) showed a soft, dynamically applied FOV vignette reduces discomfort in VR locomotion without measurable presence loss. This is the canonical demonstration that peripheral occlusion is tolerable when it is soft and situationally applied. But the manner of restriction matters greatly. Norouzi et al. (2018) found that vignetting *coupled to the user's own head motion*, strengthening as the head turns, actually **increased** sickness relative to no vignette: a directly actionable negative result that this system's motion policy inverts (effects fade *out* during head motion and restore during stability, Chapter 4), and the citation that justifies that inversion. Wu and Suma Rosenberg (2022) showed that masking only the high-optic-flow periphery beats symmetric restriction on presence: suppress the distracting *signal*, not the region wholesale. Cao et al. (2021) found granulated rest frames (sparse static noise grains, 1–7°, 25–75% density) outperform opaque black restrictors on search performance while preserving peripheral awareness, with 15 of 20 participants preferring grains to blackout. Barhorst-Cates et al. (2016) contribute the sobering datum on the far end: spatial learning survives restriction down to 10° FOV, but anxiety is elevated in *all* restricted conditions and rises monotonically as the field narrows. Waldner et al. (2014) map the comfort envelope for the flicker channel ($\approx \frac{1}{4}$ luminance range below 2 Hz achieves ≥ 0.9 detection with low annoyance), parameters this project adopts for its detection-highlighting pulse rather than anything in the 15–25 Hz photosensitivity band.

Collectively this literature draws the corridor the present system must fly through: peripheral attenuation is tolerable and even beneficial (Fernandes & Feiner), *if* it is soft-edged, decoupled from head motion (Norouzi), ideally selective about what it suppresses (Wu; Cao), and mindful that total blackout carries a measurable anxiety cost (Barhorst-Cates) and an attention- tunnelling risk (Biocca). Those constraints are visible in the final system as the dark modes' graduated opacity ceilings, every mode's motion-triggered suppression, and the evaluation's insistence on measuring peripheral awareness (H2b) and comfort (VRSQ, ratings) alongside benefit. One honest gap remains, inherited rather than resolved: no study to date measures the comfort of peripheral *desaturation* specifically over sessions longer than ten minutes, and no head-to-head of blur versus black vignette exists within a single design. This dissertation's end-of-session questionnaire collects first subjective data on adjacent questions but does not close either gap.

2.5 Platform Constraints and Prior Systems on Consumer Passthrough Hardware

2.5.1 The constraint landscape

Everything in Sections 2.2–2.3 presumes the ability to manipulate what the user sees of reality. On consumer hardware in 2026, that ability is narrowly rationed, and the rationing defines this dissertation's technical contribution (Chapter 3 develops it as a three-tier design space).

The shape of that rationing is the same across vendors. On Meta Quest 3 the operating system owns the passthrough layer, so applications can remap its colour globally but cannot blur it, mask it per-pixel, or confine an effect to a world-locked region. Passthrough Camera Access, which arrived in early 2025, grants the raw forward camera feed and full pixel control, but of a stream markedly inferior to the system layer in resolution, field of view, and latency. Apple Vision Pro is stricter, with camera pixels behind an enterprise-only entitlement and a single global surroundings dimmer for consumer apps. Chapter 3 develops the consequences as a three-tier taxonomy.

Two literature points attach to that landscape. The one published feasibility study of PCA-based on-device compositing projects 720p at 30 fps with thermal throttling within five to ten minutes, which is a simulation-based estimate rather than a measured run, and its numbers function as a warning about sustained camera processing (Laghari et al., 2025). And Varjo's research headsets grant full camera control, yet no published study was found that shader-processes their passthrough region-selectively for attention. Researchers with the capability have not asked this question, while the consumer platforms that provoke it lack the capability.

2.5.2 Prior systems and near-precedents

Every constituent mechanic of this dissertation's architecture has a documented precedent, and none of the precedents combine them. Camera-feed-on-geometry rendering with GPU effects exists in community samples. QuestCameraKit demonstrates PCA-fed materials with blur and distortion shaders applied to virtual panels and objects (Coviello, 2025), but always as content *within* the scene, never as a full-surround treated periphery. A head-centred sphere whose shader-controlled transparency reveals system passthrough exists as Meta's own "Passthrough Transitioning" motif, but the sphere carries virtual content for scene fades, not a processed camera feed (Meta MR Motifs documentation). Alpha punch-through to the passthrough underlay is a documented standard technique ("Passthrough Windows"), shipped commercially by Immersed's Passthrough Portals in 2022 so desk workers could see their keyboards, but the hole is always surrounded by *virtual* workspace, not by re-rendered reality. A peripheral dimming vignette shipped in Quest OS v67 as "Theatre View", but it explicitly does not operate over passthrough, only in immersive contexts. At the claim level, two patents describe adjacent ideas, attenuating external stimuli while preserving spatial awareness (Patents US 12548271 B2 and US 12524072 B2, 2026) and blurred external video feeds composited into VR (US 12524072), with no evidence either was implemented.

A structured prior-art search conducted for this dissertation (5 July 2026; 26 web queries and 4 GitHub API queries across ISMAR, CHI, UIST, IEEE VR, DIS and arXiv 2022–2026, Meta developer documentation, all 136 public forks of the PCA samples repository, XR trade press, and platform API documentation; the full query protocol is archived in the project record) found **no publication, product, or open-source project that composites system passthrough with a live shader-processed camera feed in a single view, for any purpose**, and none that implements peripheral passthrough manipulation for attention guidance on a standalone consumer headset. The nearest systems are the five named above, each contributing one ingredient. The hybrid composite itself, native-quality passthrough revealed through a world-locked window in an effect-processed camera-feed sphere, exploiting the quality and field-of-view asymmetry between the two layers, appears to be novel, and Chapter 3 presents it as such, scoped precisely to that composite rather than to "DR on passthrough" in general.

2.5.3 The perceptual ceiling of passthrough

One further platform datum constrains not the system but its *evaluation*: psychophysical measurement shows that no current video see-through headset reaches normal human acuity at any light level, with Quest 3 significantly worse in low light (Wang et al., 2026). Any evaluation task that requires reading fine real-world detail through passthrough therefore confounds the manipulation under test with the medium's acuity ceiling. This finding drives two decisions in Chapter 6: task stimuli are either virtual or pilot-verified legible at large print, and room lighting is bright and constant across sessions.

2.6 Attention Guidance in Applied Contexts

2.6.1 Driving-adjacent contexts

The densest applied literature on guiding attention under peripheral load is automotive, and one result from it matters structurally here. AR cueing of roadside hazards improves response time *without* degrading detection of non-target objects (Rusch et al., 2013, N = 27). That is an early empirical demonstration that guidance benefit and situational-awareness cost are separable, which is exactly the separation this dissertation's H2a/H2b pair formalises. The DR-specific studies of Murph et al. (2021) and McLaughlin et al. (2025), reviewed in Section 2.3, translate the same benefit–risk structure into diminishment terms.

A scoping statement is required here, because this dissertation uses driving *footage* as one of its stimulus contexts (Chapter 6). Per the claim-scoping rules of Chapter 1, **no claims about driving, drivers, or road safety are made anywhere in this dissertation**. The automotive literature above motivates the *scenario aesthetics*, a dynamic, high-event-rate monitoring stimulus with colour-coded semantic anchors (traffic signals), and supplies the salience-engineering precedent for ColorPop's red/orange/green hierarchy. A seated participant passively watching passenger-perspective footage licenses conclusions about attention mechanics under dynamic stimulation, nothing more.

2.6.2 Workstation and desk contexts

The workstation scenario, this dissertation's primary evaluation context, has its clearest ancestor in FocalSpace (Yao et al., 2013; Section 2.3), which established that suppressing background distraction in a mediated view improves retention of foreground content. The recent VNC-adjacent line of mixed-reality workspace studies (Section 2.3.2) brings the same question into headsets with sustained-attention and working-memory batteries. What was missing until very recently was the distractor-cost baseline in immersive settings. That has now been supplied. Peripheral visual distractors in a virtual classroom significantly increase both commission errors ($1.33 \rightarrow 3.15$, $p < .001$) and omission errors ($0.14 \rightarrow 1.18$, $p < .001$) on a Go/No-go continuous performance task ($N = 66$, within-subject), while leaving mean reaction time unchanged (Ai et al., 2025). That single result does double duty for this dissertation: it establishes that *errors, not latency*, are the sensitive currency of distraction cost (fixing the primary dependent variable of Chapter 6), and it quantifies the effect the vignette must reduce for the primary hypothesis to be supported.

2.7 Evaluating Attention Guidance in XR

2.7.1 Design and sample-size norms

The evaluation norms of this field are within-subject designs with modest samples and many repeated measures. Caine (2016)'s census of CHI 2014 found the modal sample size across all user studies to be twelve, with in-person studies smaller than remote ones. The nearest-neighbour DR studies ran $N = 16$ and $N = 12$ (Cheng et al., 2022) and two within-subject experiments analysed with mixed-effects models (McLaughlin et al., 2025). Sutton et al. (2022) ran $N = 20$ within-subject with semi-randomised condition order. The methodological lesson codified in Chapter 6 is that small-N XR studies derive their power from paired comparisons and trial-level repetition, not headcount, and that trial-level linear mixed-effects models (McLaughlin's choice) extract more sensitivity from the same sessions than aggregate t-tests. For non-normal aggregates the HCI norm has been the aligned rank transform (Wobbrock et al., 2011), though recent critique of ART's error properties (Tsandilas and Casiez, 2024) motivates this dissertation's preference for mixed models with Wilcoxon fall-backs, and Friedman tests with Bonferroni-corrected pairwise Wilcoxon comparisons for Likert batteries (the Cheng pattern).

2.7.2 Objective measures without eye tracking

Quest 3 carries no eye tracker, so this dissertation's measures must come from behaviour and head telemetry. The literature validates both channels. Reaction-time probe detection is standardised as the Detection Response Task (ISO, 2016), where button-press latency to scheduled probe stimuli indexes spare attentional capacity. That lineage motivated the driving-scene block of Chapter 6, though the task built there is simpler. Participants click on real traffic-light and pedestrian markers as they appear in the footage rather than on scripted abstract probes, and onset time, position, hit, miss, and false alarm are still machine-timed ground truth. Error rates under distraction are the sensitive measure for sustained-attention tasks (the Ai et al. result of Section 2.6.2: errors moved, latency did not). Head orientation is a validated proxy for gaze at the eccentricities that matter here: Higgins et al. (2022) formally validated head pose as a gaze surrogate for object-level attribution in VR, and Sitzmann et al. (2018) showed with concurrent head and eye tracking in 360° environments that fixations concentrate at low head velocity. Head yaw at rest is a reliable pointer to attention beyond roughly 15° eccentricity, with eye tracking adding value only inside the central window. Memory probes for attended versus unattended content (recognition of items placed at target versus periphery) complete the eye-tracker-free toolkit.

2.7.3 Subjective instruments and safety

The standard subjective battery is stable across this literature. Workload is measured with the NASA-TLX (Hart and Staveland, 1988). Simulator sickness is measured with the nine-item VRSQ (Kim et al., 2018), which suits short mixed-reality sessions, and it must be administered before *and* after exposure, because post-only scores are uninterpretable without a baseline (Brown et al., 2022). Of this battery the present study adopts the VRSQ, pre and post, and adds a single end-of-session opinion questionnaire of its own. It administers no per-condition workload instrument, and Chapter 6 records that decision and what it costs. Equivalence claims, this dissertation's "no meaningful loss of peripheral awareness" hypothesis, require dedicated machinery: the two-one-sided-tests procedure with a pre-registered margin (Lakens, 2017), which converts a would-be null result into a falsifiable claim.

2.8 Synthesis: Research Gaps and Position

2.8.1 Gaps

Read together, the eight bodies of literature above leave five identifiable gaps, three of which this dissertation addresses in whole or part:

#	Gap	Status in this dissertation
1	**No unified multi-mode DR system.** Guidance techniques and diminishment effects have been studied singly. No prior work combines multiple DR guidance modes in one configurable system for comparison within one design.	Addressed: ten modes built in one configurable system, spanning the design-space tiers of Chapter 3, of which two, SignPop and Hard Dark, are carried to confirmatory comparison within one study (Chapter 6).
2	**No hybrid of system passthrough and app-processed camera feed in one view.** Colour-only global restyling of system passthrough exists (Meta Styling API). Full pixel control of an inferior camera feed exists (PCA). No prior work composites the two, and no prior work implements peripheral passthrough manipulation for attention on a standalone consumer HMD (documented search, Section 2.5.2).	Addressed: the window-as-absence hybrid architecture (Chapters 3–4), the dissertation's strongest and most precisely scoped novelty claim.

#	Gap	Status in this dissertation
3	**No temporal transition mechanics.** Gradual formation, easing, and motion-contingent suppression of DR effects have not been formally parameterised or studied, despite adjacent evidence that ramps evade noticeability (Hata et al., 2016) and that head-coupled strengthening harms comfort (Norouzi et al., 2018).	Partially addressed: formation ramps and inverted motion-coupling are implemented and parameterised (Chapter 4) and grounded in the cited evidence. Their isolated evaluation remains future work.
4	**No user-controlled DR intensity.** User-initiated, graded release and re-application of diminishment is unstudied.	Not addressed, groundwork only (the trigger-painted window gives users spatial but not intensity control). Future work.
5	**No dual-channel DR technique.** Desaturation and blur have been studied separately, never combined and parameterised as a compound diminishment channel in AR attention guidance.	Not addressed as a confirmatory question: the compound channel exists inside ColorPop, but the study design deliberately tests the *unconfounded* dark modes confirmatorily and leaves channel decomposition to future work.

2.8.2 Position against the nearest neighbours

Two published studies stand closest to this dissertation, and the delta from each defines its contribution. **Cheng et al. (2022)** established what users want from diminishment, but inside VR simulations, because deployable DR did not exist. Their study is a perception-and-preference study of imagined systems. **McLaughlin et al. (2025)** established that distractor attenuation improves task performance, but likewise inside a virtual task environment, with the diminishment simulated by the VR scene itself. Both are simulation studies *of necessity.* This dissertation contributes the piece both were missing: a working diminishment system on real consumer passthrough hardware (Chapters 3–5), evaluated with a design that deliberately pairs a real-hardware block (the Hard Dark workstation task, with physical distractors in the participant's actual room) against a simulation block (the SignPop driving scene, a detection-gated peripheral re-grading, under oracle perception) within the same participants (Chapter 6). If the two blocks agree in direction, the field learns that the simulation methodology of Cheng and McLaughlin predicts real-hardware outcomes. If they diverge, the divergence itself is a finding about what simulation misses. Either way, the evaluation inherits the field's strongest methodological norms: within-subject design, trial-level mixed models, pre-registered directional and equivalence hypotheses, and a decision table that admits failure, assembled here, per the review above, for the first time around a deployable diminished-reality attention system.

Chapter 3

A Design Space for Diminished Reality under Compositing Constraints

3.1 Introduction

Chapter 2 established that diminished reality (DR), the selective removal or attenuation of real-world visual information (Mori et al., 2017), has a substantial conceptual literature but a thin implementation record on consumer hardware. This chapter addresses the technical half of the dissertation's spine question: *what degree of diminishment control is actually achievable on a consumer video-see-through (VST) headset, without operating-system-level access to the passthrough layer?*

The answer is not a single number but a structured design space. The central claim of this chapter is that on a platform such as the Meta Quest 3, the feasibility of any DR effect is determined almost entirely by **which layer of the compositing stack an application is permitted to touch**, and that these permissions stratify into three discrete tiers with sharply different capabilities and costs. This dissertation carries two vignette modes into the user study, SignPop and Hard Dark, both to confirmatory testing. A third, Soft Dark, occupies the midpoint of the attenuation axis but was never shown to a participant (Chapter 6). These were not designed as arbitrary aesthetic variants. Each occupies a distinct position in this space, and together they span it. The taxonomy presented here is Contribution C1, and the modes that instantiate it are described in implementation detail in Chapter 4.

The chapter proceeds as follows. Section 3.2 states the compositing constraint precisely. Section 3.3 develops the three-tier access taxonomy, the chapter's centrepiece. Section 3.4 makes explicit what is *impossible*, and why the absence itself is a finding. Section 3.5 derives three design axes from the taxonomy and locates the three modes on them. Section 3.6 states the novelty position against prior art, and Section 3.7 states, before an examiner needs to ask, what does and does not transfer to optical see-through (OST) hardware.

3.2 The Compositing Constraint

On Quest 3, the reconstructed view of the real world that the user experiences ("system passthrough") is produced and owned by the operating system. An application **cannot read, modify, or subtract from the passthrough layer. It can only composite content on top of it**. The passthrough image is never exposed to application shaders as a texture: it is composited *behind* application content by the OS compositor, at a stage of the pipeline that applications do not reach. Consequently, the naive formulation of subtractive DR, "sample the passthrough pixel, darken it, write it back", is not merely difficult on this platform. It is architecturally impossible.

This is not an incidental limitation but a deliberate privacy and safety boundary, and it is shared in stronger form by the other major consumer platform: on Apple Vision Pro, camera pixels are unavailable to consumer applications entirely (raw camera access requires an enterprise-only entitlement incompatible with App Store distribution), and the only world-appearance control offered is a global, non-spatial surroundings dimming effect (Apple, 2024). On true optical see-through glasses the constraint is harsher still and physical rather than administrative: an additive display can only *add* light to the optical path. It can superimpose brightness but cannot remove photons arriving from the world. Darkening is physically unavailable without auxiliary hardware such as per-pixel dimming layers (Itoh et al., 2021).

It is worth being clear about *why* the boundary exists, because its motivation predicts its persistence. The passthrough image is a continuous camera view of the user's home, workplace, and bystanders. Granting arbitrary applications read access to it is a privacy exposure, and granting write access is a safety exposure, since an application could imperceptibly alter the user's view of the physical world they are walking through. Both concerns strengthen rather than weaken over time, so research that assumes future read-modify-write access to system passthrough is betting against the platform vendors' incentives. This dissertation makes the opposite bet: it treats the constraint as permanent and asks what can be built *inside* it. That is what gives the resulting design space its claim to durability.

Subtractive attention guidance, dimming, muting, or blacking out the visual periphery, must therefore be *synthesized* on consumer VST hardware through whatever compositing rights the platform does grant. Characterising those rights, and the DR capability each one supports, is the purpose of the taxonomy that follows.

Diagram of the Quest 3 compositing stack: OS passthrough layer at the back (inaccessible), application render layer(s) in front, compositor combining them. Annotate the three access tiers with arrows showing what each tier can touch.

Figure 3.1: The Quest 3 compositing stack and the three tiers of access.

3.3 The Three-Tier Access Taxonomy

The Quest 3 grants applications three distinct kinds of access relevant to DR, summarised in Table 3.1 and elaborated below.

Table 3.1: Diminished-reality capability by compositing-access tier on Quest 3.

Tier	**Access mechanism**	**Achievable DR**	**Ceiling / cost**	**Modes located here**
1: OS passthrough styling	`OVRPassthroughLayer` Styling API: brightness / contrast / saturation scalars, 3D colour look-up tables, edge tint	**Global colour remapping only.** Whole-layer desaturation, tinting, posterisation	No blur. No per-pixel or spatial masks. No world-locked regions. Layer can never be read. Surface-projected passthrough (per-surface styling) deprecated at SDK v83	Not applicable (bounding case)

Tier	**Access mechanism**	**Achievable DR**	**Ceiling / cost**	**Modes located here**
2: Overlay compositing	Application draws alpha-blended geometry in front of the passthrough layer	Dimming, blackout, occlusion. A shaped *absence* of overlay passes OS-native passthrough through untouched	Can only attenuate or occlude reality. Cannot recolour, blur, or filter it	Soft Dark, Hard Dark, Granulated-Periphery, SpotLift. The focus window itself
3: Camera re-render (PCA)	Passthrough Camera Access: the raw forward camera feed as a GPU texture, with intrinsics and pose	Arbitrary per-pixel manipulation: blur, desaturation, salience re-grading, glare compression	Mono 1280×960 at ∼85–90° horizontal FOV vs the ∼110° OS view. Added latency. Binocular-fusion burden (Chapter 4). Sustained on-device processing is thermally bounded (Section 3.3.3)	SignPop (evaluated, built on the ColorPop pipeline), ColorPop, Blur (superseded), ChromaticCool, ConspicuitySqueeze, Outlined-Dark

3.3.1 Tier 1: the styling ceiling

Tier 1 deserves precision because it prevents an overclaim this dissertation must not make. It is *not* true that "nothing can be done" to the system passthrough layer: Meta's Passthrough Styling API allows an application to apply brightness, contrast and saturation adjustments, full 3D colour look-up tables, and edge tinting to the passthrough layer as a whole (Meta, 024a). A global desaturation of the entire visual field, one conceivable DR primitive, is therefore achievable at Tier 1 without touching a camera pixel.

What Tier 1 categorically lacks is **spatial control**. Every styling operation applies to the whole layer uniformly. There is no mechanism for per-pixel masks, gradients, world-locked regions, or any effect that treats one part of the visual field differently from another.

The one historical exception, surface-projected passthrough, allowed passthrough to be projected onto application-defined geometry with per-surface styling. It was deprecated at SDK v83 and is unavailable going forward. Attention guidance is *definitionally* spatial: its entire purpose is to treat the focus region differently from the periphery. So Tier 1 alone cannot implement any mode in this dissertation. **The absence of spatial control, not the absence of control, is the finding.**

3.3.2 Tier 2: diminishment by occlusion

Tier 2 is ordinary alpha compositing: the application draws translucent or opaque geometry in front of the passthrough layer. Because the VST display is an opaque screen whose every pixel the compositor ultimately controls, an application overlay can darken the apparent world without limit, up to and including full blackout, something physically impossible on additive OST optics. The constraint runs the other way: a Tier-2 overlay knows nothing about the pixels behind it. It can attenuate or occlude reality but cannot recolour, sharpen, blur, or respond to its content.

Two of this dissertation's modes live entirely at Tier 2. **Soft Dark** composites a uniform black overlay at up to 75 % opacity across the periphery. **Hard Dark** takes the same overlay to full opacity. Both leave a shaped hole, the world-locked focus window, through which the untouched OS passthrough remains visible.

That hole is the load-bearing architectural insight of the whole system, and it is worth stating in its general form: **at Tier 2, the highest-fidelity rendering of reality is an absence of rendering.** Inside the focus window nothing is drawn at all (overlay alpha = 0), so the user receives OS passthrough at its full native quality, full $\sim 110°$ field of view, and zero added latency, precisely where the user is being asked to direct their attention. The manipulation is confined to the periphery, where visual acuity is lowest and quality loss matters least. This inverts the naive design of re-rendering the whole view from the camera feed and accepting camera-grade quality everywhere, and it is why the system escapes the passthrough-acuity ceiling exactly where task performance depends on seeing well. Alpha punch-through as a mechanism is documented platform practice (Meta, 024b) and shipped commercially in Immersed's Passthrough Portals (Immersed, 2022), but in all found prior uses the hole reveals passthrough *surrounded by virtual content*. Using the hole as the *focus region of a diminished periphery* is, per the prior-art search of Section 3.6, unprecedented.

3.3.3 Tier 3: full pixel control at a price

Tier 3 is the Passthrough Camera Access (PCA) API (Meta, 2025), which since early 2025 exposes the headset's forward RGB camera feed to applications as a GPU texture together with camera intrinsics and pose. At Tier 3 the application is no longer manipulating a proxy for reality. It holds the camera pixels themselves and can apply arbitrary per-pixel computation: Gaussian blur, luminance- weighted desaturation, hue-selective salience re-grading, glare compression, everything the ColorPop mode does (Chapter 4).

The price is paid in four currencies:

1. **Resolution and field of view.** The accessible feed is a single mono stream at 1280×960 covering roughly 85–90° horizontally, against the OS passthrough's ∼110° stereo view. A Tier-3 re-render is *categorically* lower-fidelity than what the user would otherwise see.

2. **Latency.** The application-side pipeline (camera capture → texture upload → shader → display) adds delay that the OS passthrough path, with its dedicated reprojection, does not exhibit.

3. **Binocular fusion.** The feed is monocular. Presenting it convincingly to two eyes is non-trivial, and the naive approach measurably fails (the double-vision failure and its world-direction sampling solution are detailed in Chapter 4).

4. **Thermal budget.** Sustained on-device processing of the camera stream is bounded. The only published work to quantify this class of hardware for PCA-style live compositing is a simulation-based feasibility study, which projects 720p30 operation with thermal throttling after 5–10 minutes (Laghari et al., 2025), a projected, not measured, figure, since no benchmark ran on physical hardware. Tier-3 modes are, on today's hardware, modes for *sessions*, not workdays.

3.3.4 The hybrid: composing Tier 2 and Tier 3

The tiers are not mutually exclusive, and the most capable configuration this work demonstrates is a composite: the PCA camera feed, shader-processed, rendered onto a head-centred sphere as the periphery (Tier 3), with the world-locked focus window left transparent so that native OS passthrough shows through the hole (Tier 2's absence trick). The user experiences full-fidelity reality inside the window, seamlessly surrounded by a *re-graded interpretation* of reality outside it. This exploits the quality/FOV asymmetry between the two layers rather than suffering it. The prior-art search (Section 3.6) found no publication, product, or open-source project that composites system passthrough with a live shader-processed camera feed in one view, for any purpose. The implementation is presented in Chapter 4. Its planned quantitative characterisation, legibility across the two paths, latency, thermal envelope, is the subject of Chapter 5.

The hybrid composite, exploded-layer diagram: OS passthrough at the back, the effect-processed camera-feed sphere in front with the focus window cut out, and the fused view the user sees.

Figure 3.2: The hybrid composite, shown as exploded layers.

3.4 What Is Impossible, and Why It Matters

A design space is defined as much by its walls as by its rooms. Three impossibilities structure everything above:

- **No read access, at any tier, to the pixels the user actually sees.** Even Tier 3 supplies a *different, lower-grade copy* of reality, not the passthrough image itself. Any effect that must respond to what the user sees (content-adaptive dimming, saliency-weighted attenuation as proposed by Lu et al., 2012) can respond only to the camera proxy.
- **No spatial modulation of the native layer.** The full-quality view can be revealed or hidden (Tier 2) but never *partially processed*. A "blur the real passthrough periphery" mode, arguably the perceptually gentlest DR primitive per the foveated-rendering literature (Patney et al., 2016), cannot exist on this platform except via the Tier-3 copy.
- **No subtraction on OST.** The entire Tier-2 family is VST-specific (Section 3.7).

These walls explain the system's shape. The dark modes exist because attenuation-by-overlay is the only manipulation available at native quality. ColorPop exists because re-grading requires pixels, and pixels are only available as the Tier-3 copy. The window-as-absence exists because it is the sole mechanism that delivers native fidelity *and* spatial selectivity simultaneously. Had the platform offered read-modify-write access to the passthrough layer, none of this architecture would be necessary. That counterfactual is precisely what makes the taxonomy transferable: any future platform can be located in it by asking which of the three walls it removes.

3.5 Design Axes and the Placement of the Modes

Three trade-off axes fall out of the taxonomy, and the three study modes were chosen to span them.

Fidelity versus control. Tier 2 offers native fidelity with a single degree of freedom, attenuation. Tier 3 offers unlimited control at camera-copy fidelity. Hard Dark and Soft Dark sit at the fidelity pole: the periphery they show, when they show it, is real passthrough, merely darkened. SignPop sits at the control pole: its periphery is a synthetic re-grading of the camera copy, built on the ColorPop pipeline.

Suppression versus awareness. Hard Dark removes the peripheral signal entirely: maximal distractor suppression, zero residual awareness. Soft Dark attenuates to ∼75 % opacity: peripheral events remain detectable through the veil. SignPop suppresses *selectively*: the underlying ColorPop pipeline dims the red/orange/green signal band less than other colours by default, and SignPop's detection gate lifts that band to full visibility, plus a detection aperture, saliency lift and darkened surround, once a signal lamp or sign is confirmed (Chapter 4, Section 4.5.5). The user-study safety hypothesis H2b (Chapter 6) lives on exactly this axis. The axis matters because the literature puts a real cost at its far end. Users asked what they want from diminishment prefer partial attenuation over complete removal (Cheng et al., 2022), and studies of DR in working contexts find the focus gain paired with a situational-awareness risk (Murph et al., 2021; McLaughlin et al., 2025). That is why the family keeps a graded point between untouched vision and full blackout, and why the evaluation measures peripheral awareness rather than assuming it survives.

Deployability versus capability. Soft Dark and Hard Dark run without the camera pipeline at all, since their shader path samples no texture. They draw negligible GPU power, require no camera permission, and face no thermal ceiling, so they are deployable for hour-long sessions today. SignPop inherits Tier 3's full cost structure.

Together, off → Soft Dark → Hard Dark forms a clean single-variable intensity ladder (overlay alpha 0 → 0.75 → 1.0), while SignPop extends the space along the orthogonal control axis. This is the sense in which three modes "span" the space: two axes, both covered, with the intensity axis sampled at three points.

These three are the modes the study was designed around, though only two of them reached a participant, and they are not the whole family. Ten peripheral treatments were built in total, and the remaining seven fill in the same two axes at points the study had no budget to test. They also divide along the tier boundary, which is the cleanest evidence that the boundary is real rather than imposed. Four modes never read a camera pixel and are therefore Tier 2: the two dark modes, plus GranulatedPeriphery, whose grains come from a noise texture rather than the world, and SpotLift, whose periphery is a plain dark overlay and whose focus-window brightness lift exists only in the video path, because OS passthrough cannot be brightened. The other six read the camera feed and are therefore Tier 3: Blur, ColorPop and the SignPop that extends it, ChromaticCool, ConspicuitySqueeze, and OutlinedDark, which needs five camera taps per fragment just to find the edges it draws back in. Chapter 4 gives each of them in implementation detail. The point for the taxonomy is that the tiers were not fitted to three convenient examples. They were derived from building the whole family and finding that every member's ceiling was set by the tier it had to run at.

A fourth, orthogonal dimension deserves naming even though this dissertation treats it as design rather than as an experimental factor: **time**. Every mode has temporal behaviour: a formation transition when the window locks, a fade-out when the head moves fast, a fade-in when it settles (Chapter 4). The literature review found these transition mechanics to be essentially unstudied in DR. Prior work evaluates diminishment as a static state, not as something that arrives, yields, and returns. The system therefore exposes every temporal parameter as tunable, and the user study's exploratory interview asks specifically about the motion-fade experience. A controlled study of DR transition dynamics is identified in Chapter 7 as future work seeded by this design space rather than resolved by it.

3.6 Novelty Position

The claims of this chapter are calibrated against a structured prior-art search conducted on 2026-07-05 across ISMAR, CHI, UIST, IEEE VR, DIS and arXiv (2022–2026), Meta developer documentation, GitHub (including all 136 forks of the upstream PCA samples repository), XR industry press, and the visionOS and Varjo API surfaces; the full query protocol is archived in the project record. Three verdicts matter here:

1. **The hybrid composite** (system passthrough revealed through a window in a live shader-processed camera-feed sphere): *nothing found.* No paper, product, or open-

Two-axis plot (fidelity/control × suppression/awareness) with the three modes placed, deployability annotated as marker size or colour.

Figure 3.3: Two-axis plot (fidelity/control × suppression/awareness) with the three modes placed, deployability annotated.

source project, for any purpose.

2. **Peripheral passthrough manipulation for attention guidance on a standalone consumer HMD**: *nothing found as an implemented system.* The nearest work is concept-level or differently situated. McLaughlin et al. (2025) evaluate DR distraction-attenuation entirely inside VR simulation. The DIS 2026 ADHD co-design study is formative, with no prototype. DiminishAR and the CHI 2026 Vision Pro study target *specific objects/persons* with occlusion rather than the periphery. Sutton et al. (2022) modulate real-world saliency but on a bench-mounted optical see-through rig that can only add light.

3. **A dark peripheral vignette over passthrough**: *near precedent, no direct precedent.* Quest OS v67's "Theatre View" ships an adjustable peripheral dimming vignette, but explicitly not over passthrough (immersive mode only). Vision Pro's surroundings dimming is global, not windowed.

Each constituent mechanic, taken alone, has documented precedent that this dissertation cites rather than claims: camera-feed geometry with GPU effects (Coviello, 2025), a head-centred fade sphere revealing passthrough (Meta, 024c), alpha punch-through windows (Meta, 024b; Immersed, 2022), OS peripheral dimming in immersive mode (Meta, 024d), and saliency modulation of reality on OST hardware (Sutton et al., 2022). The defensible novelty claim is therefore precisely bounded: **the composite**, native passthrough through a world-locked focus window in an effect-processed camera-feed periphery, exploiting the quality/FOV asymmetry between layers, **applied to peripheral attention guidance on a consumer standalone headset**, together with the world-direction fusion sampling that makes the mono feed binocularly viewable (Chapter 4). An alpha-blended dark overlay, by itself, is just compositing, and is not claimed as novel.

Two adjacent patents (Patents US 12548271 B2 and US 12524072 B2, 2026) describe claim-level ideas in this territory: attenuating external stimuli while preserving spatial awareness, and blurred external video feeds composited into VR. Neither shows evidence of implementation. They are noted for completeness.

3.7 What Transfers to Optical See-Through, and What Does Not

This section makes the non-transfer statement before an examiner makes it. The Tier-2 dark overlay family, Soft Dark, Hard Dark, and the darkness component of every mode, works because a VST display is an opaque screen over which the compositor has total authority. **It does not transfer to optical see-through glasses**, whose additive optics cannot remove light from the world. On OST, "draw black" means "draw nothing". A hypothetical OST implementation of peripheral dimming would require optical hardware assistance (global or segmented dimming layers, per-pixel occlusion masks), which is exactly Tier-1-style capability relocated into the optics.

What does transfer is, first, the **taxonomy itself**. OST platforms can be located in it: their "Tier 2" is empty for subtraction, and their achievable DR collapses onto additive saliency modulation of the Sutton et al. kind, plus whatever dimming hardware exists. Second, the **human-factors findings** of Part H transfer, because what peripheral diminishment does to attention is a property of the human visual system, not of the display that implements the diminishment. The engineering is platform-bound. The psychology is not. That division of transferable from non-transferable is itself a use of the design space, and it is the frame in which Chapter 7 discusses generalisation.

3.8 Summary

On consumer VST hardware, diminished reality is not one problem but three, stratified by compositing access: a colour-only, spatially-blind styling tier, an attenuation-and-occlusion overlay tier whose most powerful move is a shaped absence, and a full-control camera-copy tier that pays in fidelity, latency, fusion and heat. The three study modes span the space this stratification creates, the hybrid composite exploits the asymmetry between its tiers, and the walls of the space, no reads, no spatial modulation of the native layer, no subtraction on OST, are findings in their own right. Chapter 4 descends from this map into the machine.

Chapter 4

System Design and Implementation

4.1 Overview

This chapter presents the reference implementation (Contribution C2): a multi-mode diminished-reality attention-guidance system running on a consumer Meta Quest 3, built in Unity 6 on Meta's Passthrough Camera Access sample framework (Meta, 2025). The system realises the design space of Chapter 3 in a deliberately economical architecture, **one shader and one manager component on a single piece of geometry**, and this chapter argues that the economy is itself a result: every capability tier of Chapter 3's taxonomy is reachable from one head-centred sphere and one fragment program.

The exposition follows the architecture outward. Section 4.2 describes the sphere-and-window geometry that everything else inhabits. Section 4.3 presents the hybrid composite of system passthrough and processed camera feed, the flagship technique. Section 4.4 explains the world-direction sampling scheme that makes a monocular camera feed binocularly fusible. Section 4.5 specifies the three study modes at implementation depth, together with the superseded and exploratory modes retained in the codebase. Sections 4.6 and 4.7 cover the motion-based comfort suppression and the interaction design. Section 4.8 describes the perception subsystem: live detection support and the offline oracle pipeline that stands in for it in evaluation. Section 4.9 describes the video test scene used as the evaluation harness. Section 4.10 is the failure museum: the documented dead ends whose lessons shaped the final design.

Throughout, parameter names in `code font` are the actual serialized fields and shader properties in the repository (`CameraSphereVignette.shader`, `CameraSphereVignetteManager.cs`, `VideoTestSceneManager.cs`), so that every quantitative claim in this chapter is checkable against the artifact.

System architecture: OVRCameraRig, the CameraSphereSphere GameObject with manager and material, PCA camera feeds, OS passthrough behind, and the compositor output. One box per component, arrows for per-frame data flow.

Figure 4.1: System architecture and per-frame data flow.

4.2 The Sphere and the Window

4.2.1 Geometry

The entire visual effect is carried by a single inverted sphere of approximately 10 m radius, centred on the user's head and re-centred every frame (translation only: the sphere never rotates, which is what keeps its surface coordinates world-stable). The mesh is rendered inside-out with `Cull Front`, on the transparent queue (`Queue = Transparent`), with depth writes disabled (`ZWrite Off`) and conventional alpha blending (`Blend SrcAlpha OneMinusSrc` Every fragment the user can see, in any direction, is a fragment of this sphere. The shader's output alpha at that fragment decides, per pixel, whether the user sees the diminishment effect or whatever lies behind the sphere, which, in the passthrough scene, is the operating system's native passthrough layer.

This choice of a *fragment-alpha-controlled full-surround surface* is what makes one object span all of Chapter 3's tiers: alpha 0 is Tier 2's "absence" (native passthrough passes through untouched); alpha 1 with a black colour is Tier 2's blackout; alpha 1 with a camera-fed colour is Tier 3's re-render. The mode logic reduces to deciding, per fragment, a colour and an alpha.

4.2.2 The world-locked focus window

This dissertation uses two different terms for two different things that can look similar: both are places where the periphery effect briefly does not apply.

The **focus window** is the region the user places and locks onto a spot in the room. It stays fixed to that spot. Inside it, the application draws nothing at all, so native passthrough shows through untouched. The user controls it directly, by painting it once, and it does not move again until they reset it.

The **detection aperture** is a different thing, described fully in Section 4.5.5. It is a smaller opening the system creates over a traffic light, sign, or person that a live detector has just recognised. It appears and disappears as objects are recognised and lost. The user has no control over it at all.

The rest of this chapter keeps these two terms separate and names whichever one it means.

The focus window is stored not in screen space or head space but in **world spherical coordinates**: a rectangle `_FocusRect = (azMin, azMax, elMin, elMax)` in radians of azimuth and elevation about the sphere's centre. Each fragment reconstructs its own direction from first principles:

```
dir = normalize(worldPos - _SphereCenter)
az  = atan2(dir.x, dir.z)
el  = asin(dir.y)
```

and computes a signed distance outside the rectangle, `max(dAz, dEl)`, which a `smoothstep(0, _S` converts into the per-fragment vignette coordinate `t`: 0 inside the window, rising smoothly through the soft edge (default `m_softEdgeDeg` $= 20°$, widened to 32° for ColorPop, Section 4.5.2), and 1 in the far periphery. Every mode consumes this single scalar. The dark modes use it directly as overlay alpha. ColorPop uses it to interpolate each pixel between its "inside" and "outside" treatments.

The 20° default is chosen against the useful field of view, the region within roughly 30° of fixation from which people actually extract task-relevant information without moving their eyes or head (Ball and Owsley, 1993). Placing the soft edge inside that radius means the transition band falls where the eye is still resolving detail, so the effect reaches full strength just as the useful field runs out, rather than cutting sharply across it. It also sits above the $\approx$15° eccentricity beyond which head yaw becomes a reliable pointer to attention (Higgins et al., 2022), which matters because the window is placed by head-directed painting and must be larger than the noise floor of the pointing method that places it.

Because azimuth and elevation are computed from world direction, the window is **world-locked**: it stays fixed to the room, to the desk, the doorway, the whiteboard, as the head rotates, rather than travelling with the gaze. Direction is computed per-fragment from `worldPos` rather than interpolated from vertices, which avoids interpolation error at the sphere's poles.

The az/el window geometry: sphere cross-section with the focus rectangle drawn on it, the soft-edge band, and the t-profile plotted along one axis.

Figure 4.2: The az/el window geometry.

4.2.3 The window as absence

In the passthrough scene the shader's output inside the window is `alpha = t = 0`: nothing is drawn. The consequence, argued in Chapter 3 and repeated here because it is the system's central design decision: inside the window the user sees the operating system's own passthrough at full native quality, full ∼110° field of view, and the OS pipeline's own latency compensation, none of which any application-rendered reconstruction can match. The effect budget is spent entirely in the periphery, where human acuity is lowest. One line of shader arithmetic (`alpha = t`) simultaneously solves the field-of-view problem (the accessible camera feed covers only ∼85–90°) and the fidelity problem (1280×960 mono versus the native stereo view), by ensuring the reconstruction is only ever shown where neither limitation matters much.

4.2.4 Per-frame data flow

The manager component (`CameraSphereVignetteManager`) owns all per-frame state and feeds the shader exclusively through material uniforms, so the shader holds no state of its own. Each frame it re-centres the sphere on the head anchor and uploads `_SphereCenter`, the one value the whole angular coordinate system depends on, uploads the focus rectangle and soft edge, combines formation and motion-suppression state into the single `_VignetteStrength` scalar, and pushes the active mode's parameter block (Section 4.5). For camera-fed modes it also uploads each active camera's pose basis and half-FOV tangents, the latter derived from the intrinsics the PCA API reports (`SensorResolution / (2*FocalLen` rather than an assumed field of view, which is what keeps the camera periphery in scale with the native window across the seam.

Stereo correctness is handled at the vertex and fragment interface. The shader is written for Unity's single-pass instanced stereo and, by the world-direction design of Section 4.4, contains *no* eye-dependent sampling anywhere in the camera path, so the renderer needs no per-eye scripting. The selection dots' material is a serialized template asset rather than a runtime-found shader, for the reason given as failure 3 in Section 4.10.

4.3 The Hybrid Composite

The system's flagship configuration composes the tiers: a **Tier-3 processed camera feed as the periphery, with the Tier-2 transparent window revealing native passthrough as the focus region**. These are two different renderings of the same reality, fused into one continuous view.

Mechanically: the Passthrough Camera Access feed is bound to the sphere material (`_MainTexL`, and `_MainTexR` where the right camera is available), the fragment shader computes each periphery fragment's colour from the feed (with whatever manipulation the active mode specifies: blur, desaturation, salience re-grading), and outputs it at `alpha = t*strength`. Inside the window, `t = 0` yields transparency and the OS layer shows through. The seam between the two renderings is the window's soft edge, where the processed feed fades out over `_SoftEdge` radians. Because both renderings depict the same world from (nearly) the same viewpoint, the transition reads as an effect boundary, not as two images.

The construction has one honest seam-level caveat, stated here because it motivates a Chapter 5 benchmark. The two renderings do not share a latency budget: the OS passthrough inside the window is served by the platform's dedicated low-latency reprojection path, while the camera-fed periphery carries the application pipeline's additional delay. In static or slowly moving views the mismatch is invisible. Under fast head rotation the periphery can momentarily lag the window at the seam. The system's motion-based suppression (Section 4.6) incidentally masks the worst case, since the effect fades out precisely when the head moves fast, but the residual mismatch is a real property of the hybrid and is measured, not assumed away, in the planned latency benchmark.

The claim this construction supports is deliberately precise (Chapter 3, Section 3.6): the *composite* is the novelty, exploiting the quality/FOV asymmetry between a native layer that cannot be touched and a camera copy that can, not any of its ingredients. Its costs are Tier 3's costs (resolution, latency, fusion, heat), paid only in the periphery. Its planned quantitative characterisation (per-mode GPU cost, legibility of window versus periphery, thermal envelope) is specified in Chapter 5.

Through-the-lens style mock-up or screenshot pair: the same room with the hybrid active, native-quality window, processed periphery, annotated with which tier renders which region.

Figure 4.3: The hybrid composite as seen through the lens: native-quality window, processed periphery.

4.4 Binocular Fusion by World-Direction Sampling

The accessible camera feed is monocular, but the display is binocular. How the mono image is presented to two eyes determines whether the system is usable at all. The governing requirement is that the sampling domain be **eye-independent**. Under stereo rendering the same world point projects to different screen positions in each eye, so any scheme keyed to screen position feeds each eye a different camera pixel for the same world point, and the two images do not fuse (Section 4.10, failure 1).

The shader therefore samples by **world direction**. Each fragment's direction from the sphere centre (already computed for the window test) is projected through a head-centred pinhole model of the camera: with the camera's forward/right/up basis (`_CamLFwd`, `_CamLRt`, `_CamLUp`) and half-FOV tangents (`_TanHalfFovL`, derived from the camera intrinsics as `SensorResolution / (2*FocalLength)` rather than an assumed FOV), the shader computes

```
uv = ( (dir.rt, dir.up) / max(dir.fwd, eps) ) / tanHalfFov . 0.5 + 0.5
```

Because world direction is **eye-independent**, both eyes sample the *same* camera pixel for the same world point. The mono image behaves like a texture painted onto the world shell, and the visual system fuses it. The residual, and acceptable, artefact is that the painted shell has no true stereo disparity of its own. This is consistent with the project's standing comfort constraint that a mono feed must never be presented as if it were geometry-correct stereo across the full field of view.

Where both cameras are available, the shader samples both and selects per fragment by in-view weight, a **hard split** (`inFovL >= inFovR ? sampledL : sampledR`) rather than an alpha crossfade, because a crossfade of two horizontally offset viewpoints produces exactly the ghosting the world-direction scheme exists to avoid. The two forward cameras share an orientation and differ only by a small horizontal baseline, so the stitch line is invisible in normal use.

Fusion diagram: same world point, two eyes. Screen-space sampling (two different camera pixels, double image) versus world-direction sampling (one camera pixel, fused).

Figure 4.4: Screen-space versus world-direction sampling of a mono camera feed.

4.5 The Modes

The mode enumeration in the codebase (`VignetteMode`) contains eleven entries. Two are carried into the user study, SignPop and Hard Dark, both to confirmatory testing. A third, Soft Dark, is specified here but was never tested with participants (Chapter 6). Blur is their superseded ancestor. ColorPop is the base grading pipeline SignPop extends with a detection gate (Section 4.5.5); it is not itself evaluated as a separate mode. Five further entries are retained design-exploration variants (ChromaticCool, ConspicuitySqueeze, GranulatedPeriphery, OutlinedDark, SpotLift). The eleventh, TintedDark, is dead code, dropped from discussion but kept in the enum because removing the entry would break a serialized scene value (Section 4.10). All share the geometry, window, and t-coordinate of Section 4.2. A mode is, in implementation terms, a per-fragment colour/alpha policy.

4.5.1 Peripheral occlusion (Hard Dark and Soft Dark)

The simplest and strongest manipulation: `periColor = black`, `periAlpha = t * _VignetteStreng` with `_MaxVignetteAlpha = 1.0`. At full formation the periphery is completely occluded. The user's visual world *is* the focus window. The shader path (`_SimpleMode = 1`) samples no textures and does no camera mathematics. An early return delivers what is effectively a fixed-function overlay, with correspondingly negligible GPU cost and no camera permission or thermal exposure (this is the deployability argument of Chapter 3, Section 3.5, in code). Hard Dark is the confirmatory mode of user-study Block A (Chapter 6): the strongest available manipulation, chosen to maximise the chance of detecting the distraction- suppression mechanism if it exists. The same shader path, run with `_MaxVignetteAlpha` set from `m_mode2MaxAlpha` (default **0.75**) instead of 1.0, gives Soft Dark: the periphery darkens substantially but never fully, so peripheral motion and events stay visible through the veil. The ceiling is set below 1.0 rather than at it because Cheng et al. (2022) found users asked what they want from diminishment prefer partial opacity reduction to complete removal, and want enough residual structure to know something is still there. Soft Dark is that preference implemented as a number. It is one of the further design-space variants of Section 4.5.3, and it carries no participant data (Chapter 6).

Both dark modes **form gradually**: on window lock, a coroutine animates `_VignetteStrength` from 0 to 1 along a SmoothStep over `m_vignetteFormTime` (default **3 s**), so the periphery dims as a transition rather than a cut. The ramp is there because Hata et al. (2016) showed that a visual change introduced gradually enough stays below the awareness threshold while still biasing gaze, so the guidance survives the ramp even though the announcement does not. Formation runs independently of the motion suppression of Section 4.6. If the user's head moves during formation, suppression multiplies the strength down while the formation clock keeps counting, and the effect resumes at its accumulated level when the head settles.

4.5.2 Colour- and saliency-gated peripheral re-grading (ColorPop)

ColorPop is the Tier-3 mode. It re-grades the camera feed's colour and brightness rather than dimming it, using the literature's strongest guidance channel: luminance and saturation contrast, with no hue shifts (Bailey et al., 2009; Grogorick et al., 2017; Sutton et al., 2022). The re-grading is a standing property of the whole visual field, not a transient cue that appears and disappears. After on-device comparison across the mode family, ColorPop was the standout and became the dissertation's dynamic-scenario mode (user-study Block B).

Its grading is a **four-level hierarchy** on two tracks, gated by colour class and interpolated by the window coordinate t:

$$\text{outside-other} < \text{outside-ROG} < \text{inside-other} < \text{inside-ROG}$$

Colour classification. A shader helper (`PopColorKeep`) computes membership in the kept red/orange/green (ROG) set, the colour classes of signal lamps and signage. Membership is defined by hue bands (warm band hue $\in$ [0, 0.18] fading out by 0.25; a wrap-around red band above 0.85; a green band 0.22–0.48) gated by saturation floors. The warm band uses a deliberately higher floor (`_PopWarmSatMin` = 0.35): warm-*white* light, headlights at hue $\approx$ 0.1 and saturation 0.1–0.25, must not ride into the "orange" class, while genuine lamps and signs sit at saturation above 0.5.

The "other" track (non-ROG pixels) stays near-natural inside the window (a slight desaturation, `_PopInsideDesat` = 0.25, keeps the window from feeling artificially vivid against its own periphery), graded down to dim grey outside (`_PopGreyDim` = 0.4). The periphery's non-signal content is present but muted.

The ROG track: inside the window, saturation is boosted $\times$`_PopSatBoost` (1.7) about the pixel's own luminance, a sigmoidal midtone-contrast transfer is applied per Sutton et al.'s published recipe ($\alpha = 10$, $\beta = 0.5$, normalised so 0$\to$0 and 1$\to$1; strength `_PopSigmoid` = 0.6, scaled by $(1 - t)$ so it is full inside the window and absent outside), and brightness is lifted $\times$`_PopBrightIn` (1.15). Outside the window, ROG pixels keep their **natural colour, dimmed** to $\times$`_PopBrightOut` (0.65). This is a deliberate design decision: signal-relevant colours remain detectable *everywhere*, merely privileged inside the window, which fits a monitoring context in which a peripheral traffic light must never become invisible.

Glare compression. Pixels that are bright (`luma > _PopGlareKnee` = 0.6), unsaturated, and not ROG, the signature of headlight glare, are compressed toward the knee, i.e. toward a visible dim spot, never black: the oncoming vehicle stays legible while its veiling glare is attenuated. The compression is mild inside the window (`_PopGlareInside` = 0.35) and strong in the periphery (`_PopGlareOutside` = 0.85). A **blown-core guard** protects the one case the mask would otherwise destroy: the clipped white centre of a bright signal lamp. Before dimming, the shader samples a small Gaussian surround (`_PopGuardRadius` = 0.008 UV). If the surround is a kept colour, the lamp's saturated fringe, the pixel is exempted.

Global periphery dim. Finally the whole periphery, ROG included, is multiplied down by `_PopPeriphDim` (0.85), so that the focus window also wins on plain luminance, the strongest single guidance channel in the attention literature (Bailey et al., 2009; Grogorick et al., 2017). Net outside-ROG brightness at defaults is therefore $\approx 0.65 \times 0.85 \approx 0.55$ of natural. That figure is a deliberate midpoint rather than a maximum. Cheng et al. (2022) found partial attenuation preferred over removal, and Barhorst-Cates et al. (2016) found anxiety rising monotonically as the visible field narrows, so a monitoring mode that must keep a peripheral traffic light findable has reason to stop at roughly half natural brightness rather than push further.

Because a colour/grey boundary reads perceptually harsher than a blur or darkness boundary, ColorPop uses a widened window edge: `m_popSoftEdgeDeg` = **32**° against the global default of 20°.

In the passthrough scene, ColorPop grades the periphery only. The window must remain transparent, since the OS layer cannot be graded, and repainting the full field from the mono feed would violate the comfort constraint. In the video test scene (`_PassthroughMode = 0`) the same shader grades the entire sphere, window included (`periAlpha = lerp(1, t, _Passthrough`
which is the configuration evaluated in Block B.

ColorPop grading pipeline flowchart: classify (ROG?) → glare mask + core guard → track-specific grading → periphery dim, with a screenshot pair showing a street scene off/on.

Figure 4.5: ColorPop grading pipeline flowchart.

4.5.3 Progressive peripheral blur with contrast restoration (Blur, superseded) and the exploratory family

The system's ancestor mode, **Blur**, renders the periphery from the camera feed with progressive Gaussian blur and delayed desaturation, and is retained in the codebase though not evaluated. It blurs through a two-ring Gaussian (centre + 8 taps at radius r + 8 taps at 2r, σ = r) whose fine/coarse difference doubles as a local Laplacian estimate driving a **perceptual noise term** (Tariq et al., 2022): world- anchored luminance noise, amplitude-scaled by local variance (capped at 0.2 to prevent marbling at hard night-scene boundaries), reintroducing "detectable but not resolvable" high-frequency content that blur otherwise strips. A **contrast-restoration** term (Patney et al., 2016) re-amplifies the blurred signal's residual local contrast (`_BlurContrastRestore` = 0.8, overshoot clamped to ±0.15), because peripheral blur is detected primarily through its contrast loss. Restoring contrast roughly doubles tolerable blur radius and removes the tunnel-vision percept.

Blur precedes nothing: desaturation begins at `_DesatDelay` $= 0.3$ of the t-range and blur only at `_BlurDelay` $= 0.5$, so colour mutes before structure softens.

Two sampling details make the blur path viable on the equirectangular video sphere as well as the camera feed. Blur taps go through a mode-aware UV helper: in camera mode both axes clamp (the camera image does not wrap), while in equirectangular mode the horizontal axis wraps (`frac`) for a seamless 360° join and out-of-range vertical taps are *folded back* rather than clamped, since clamping would smear the top and bottom texture rows across the sky and floor. Independently, the blur radius is faded to zero toward the poles by a `cos(el)` factor, since at the poles any UV-space offset corresponds to a huge angular distance and unfaded taps produce visible polar artefacts.

Five further exploratory modes are retained in shader and enumeration but excluded from evaluation. Each is a literature-derived point in the design space: warm-window, cool-periphery chromatic opposition (**ChromaticCool**); centre–surround contrast flattened toward the local mean (**ConspicuitySqueeze**, after Veas et al., 2011); world-locked static noise grains with eccentricity-ramped density (**GranulatedPeriphery**, after Cao et al., 2021); near-blackout with luminance edges restored (**OutlinedDark**, after Cheng et al., 2022's finding that context-preserving outlines reduce anxiety); and a focus-region brightness lift over a soft peripheral dim (**SpotLift**, video mode only, since OS passthrough cannot be brightened). They are reported as explored design points. None carries evaluation claims. A sixth candidate, **TintedDark** (a configurable-colour overlay), is dead code, since it has no proposing paper and no mechanism the dark family does not already have. Its shader branch and enum entry remain in the codebase, because removing the entry would break a serialized scene value. The video test scene's free-play mode cycle (`k_modeCycle`, the A-button cycle) walks the entire eleven-entry enum in enum order, matching the passthrough scene's cycle, so free play and demos can reach every mode, while the study proper only ever programmatically selects SignPop, Soft Dark, and Hard Dark (Chapter 6).

4.5.4 Cost structure of the shader paths

The mode family spans roughly two orders of magnitude in per-fragment cost, and the gradient is structural rather than incidental. It is Chapter 3's deployability axis realised in instructions. The dark modes take an early return after the window test: no texture reads, a handful of ALU operations, one blended write. ColorPop reads the source texture once per fragment on its main path, plus a 17-tap Gaussian surround *only* inside the small fraction of fragments where the glare mask fires with the core guard enabled. Its cost is dominated by per-pixel colour arithmetic (HSV-style classification, two grading tracks, a sigmoid). The Blur path is the heaviest: two Gaussian rings of 8 taps each around a shared centre (17 taps) per camera, doubled when both cameras are live near the stitch, plus the noise texture read and contrast-restoration arithmetic. These are analytic expectations. Their empirical confirmation (per-mode GPU frame cost against the 72/90 Hz budget) is one of the planned benchmarks of Chapter 5. The design consequence stands regardless of exact numbers: the modes the study asks participants to imagine using for long sessions (the dark family) are the modes that cost almost nothing, and the mode with the richest effect (ColorPop) is the one whose Tier-3 pipeline carries the platform's thermal ceiling.

Screenshot triptych of the three study modes on the same video frame.

Figure 4.6: Screenshot triptych of the three study modes on the same video frame.

4.5.5 Detection-gated salience re-grading (SignPop, video-only): evaluated, Block B

SignPop is an eleventh mode, live only in the video test scene, and it is the mode placed in front of participants in Block B (Chapter 6). It shares ColorPop's four-level ROG grading pipeline (Section 4.5.2), but a confirmed detection from the offline oracle track (Section 4.8.2) gates a bundle of effects, not colour alone.

The colour gate. ColorPop already keeps red/orange/green hues partly visible in the periphery without any detection (Section 4.5.2's outside-ROG track). SignPop restricts the *full* vivid treatment to pixels inside a confirmed detected traffic-light/stop-sign box (`_PopDetGate`). An undetected ROG pixel, neon signage, brake lights, advertising, or a genuine bake miss, falls back to a partial keep (`m_signRogFallback` / `_PopDetFallback`, default 0.35) rather than full grey, a "graceful miss" so a missed real signal is dimmed, never hidden. A short hold (`m_signDetHoldSec`, default 0.5 s) keeps a detection alive after it drops out of the bake so pops do not blink between samples.

An independent detection-zone system. Beyond the colour gate, every confirmed detection drives three further, colour-independent effects computed from soft elliptical zones around each detection box (`CameraSphereVignette.shader:610-665`):

- **A detection aperture.** The vignette is carved away directly over the object (`t = min(t, 1 - windowClear)`). This is not the focus window described in Section 4.2: it is a separate, transient opening that appears wherever a detection is confirmed, inside or outside the focus window, and disappears when the detection is lost.
- **A saliency lift.** Saturation, brightness, and local contrast around the object are boosted (`detHighlight`), scaled down when the object already sits inside the focus window (`_DetectionOutsideScale`) since the standing focus window already reveals it there, and left at full strength in the periphery where the lift is doing the work.
- **A darkened surround.** A soft annulus just outside the object's boundary is dimmed slightly (`detSurround`), raising the object's centre–surround contrast from both sides (Veas et al., 2011's dual modulation, the same source cited for ConspicuitySqueeze, Section 4.5.3).

This entire detection-zone system is suppressed for Hard Dark (`_SuppressDetectionWindows`): once the user-placed window is locked, no detected object anywhere gets a carve-out, however salient. This is a deliberate choice, not an omission: Hard Dark's role in Block A (Chapter 6) is to test the strongest possible manipulation, full occlusion with no exceptions, against a workstation task with no peripheral object the participant needs to see. Adding detection apertures back into Hard Dark would weaken exactly the property the confirmatory test relies on. A detection-aware variant of Hard Dark is a reasonable extension for contexts where a peripheral object genuinely matters, such as a walking or driving scenario, and Chapter 8 returns to it as future work, not as a gap in the present study's design.

A breathing pulse. The saliency lift additionally oscillates at approximately 1 Hz (`_DetectionPulseAmp` modulating `_Time.y`, `CameraSphereVignette.shader:909-912`), following Waldner et al. (2014) on low-frequency, low-amplitude temporal modulation attracting attention with low annoyance.

Two gates on what counts as a detection at all. The offline bake (`Tools/bake_detections.p`
considers only the forward view by default, azimuth ±85°, elevation ±50° around video-centre, so a detection behind the participant never enters the track. At runtime, a detection's window and boost stay suppressed until its apparent size clears a minimum threshold

(~1.2°, `VideoTestSceneManager.cs:200`). 33 of 246 post-debounce traffic-light lifetimes in the current bake never clear it, so a distant, still-small light does not open a window over what is still, perceptually, a speck.

What a confirmed detection actually switches on. Six effects fire together on the same gate: the colour-pop upgrade, the window carve-out, the saturation lift, the brightness lift, the contrast expansion, and the darkened surround, layered on the ~1 Hz pulse. None of the six is isolable in the current build. Chapter 6's threats-to-validity section and Chapter 7 return to this as a limitation on what any measured effect can be attributed to.

4.6 Motion-Based Suppression

Head-coupled restriction of the visual field is a known comfort hazard: Norouzi et al. (2018) found that vignetting yoked to head rotation *increased* rather than decreased simulator sickness. The system therefore decouples in the opposite direction: the effect **fades out while the head moves fast, and returns when it settles**.

Each frame the manager computes head angular speed between successive head rotations. When speed exceeds a per-mode threshold, a suppression scalar animates toward 1 over 0.20 s, and once speed has stayed below threshold for the mode's hold time, back toward 0 over 0.70 s. Thresholds and holds scale with the mode's intrusiveness: Blur 30°/s (hold 0.6 s), ColorPop 35°/s (0.7 s), Soft Dark 50°/s (1.0 s), Hard Dark 70°/s (1.5 s). The blackout mode therefore tolerates the most motion before yielding and waits longest before returning, because wrongly suppressing a strong effect flashes the whole periphery back into view, while wrongly suppressing a gentle one costs nothing. The fade asymmetry follows the same logic: yielding must feel instantaneous to be trusted, and returning may be leisurely because nothing is at stake.

The shader receives `effectiveStrength = m_vignetteStrength . (1 - suppression)`, so suppression multiplies the formation state rather than resetting it, and a glance over the shoulder during the 3-second formation does not restart the transition. State the user created, the locked window and the accumulated formation, is never destroyed by a comfort mechanism, only veiled.

One refinement matters for the interaction loop. If the head's direction enters the focus window, with a +15° margin, the hold timer is zeroed immediately, so arriving back at the focus region restores the effect at once. The asymmetry is deliberate. Looking *away* is treated as a possible need for situational awareness, so the effect yields generously, while looking *back at the task* is treated as re-engagement, so it returns promptly.

Suppression state diagram: speed threshold, hold timer, fade-out/fade-in ramps, and the focus-arrival shortcut.

Figure 4.7: Motion-based suppression: speed threshold, hold timer, and fade ramps.

4.7 Interaction Design

The user defines the focus window by **painting** it: holding the right controller trigger sweeps the controller's aim across the scene, and the window rectangle grows to enclose the swept region (seeded at the aim point with a small brush pad). Releasing the trigger **locks** the window and, in the dark modes, starts the 3-second formation. The rectangle is computed and stored directly in the world az/el coordinates of Section 4.2.2 (via `atan2`/`asin` on the controller-forward direction), so what the user paints is what stays locked to the room. The **A** button cycles modes. **B** clears the window and cancels the effect.

Feedback is deliberately minimal. Five dot markers, four corners plus an aim cursor, show the window during painting, dim on lock, and auto-hide after 3 s. A world-space toast announces mode changes and fades away after 2.8 s. Hand tracking is disabled and input is controller-only, a scope decision that removes a class of input ambiguity from the study sessions.

Three design rationales are worth making explicit. *Painting rather than pointing*: a focus window is a region, not a point, and sweeping it makes the region's extent a deliberate act, where auto-sizing a window around a pointed target would hide a consequential parameter inside a heuristic. *World-locking rather than head-locking*: the window belongs to the task surface, not the gaze, and a head-locked window would chase attention rather than anchor it, which is also the comfort hazard Section 4.6 exists to avoid. *Formation as transition*: the 3-second ramp exists because an instantaneous peripheral blackout is startling, and it doubles as the user's confirmation that the lock succeeded. These temporal behaviours are, per Chapter 2's gap analysis, among the least-studied parameters in DR design, so the system exposes every one of them as a tunable value rather than a constant.

Interaction sequence: paint (dots visible) → release/lock → formation → dots auto-hide, plus the mode toast.

Figure 4.8: Interaction sequence for placing and locking the focus window.

4.8 Perception: Detection Islands and the Offline Oracle

4.8.1 Live detection support

The live passthrough manager (`CameraSphereVignetteManager`) supports up to eight simultaneous **detection islands** (`k_maxDetections = 8`, backed by the shader's `_DetectionRects[16` array): az/el rectangles, fed by an optional `YoloRunner` component, inside which the vignette is cleared (`t = min(t, 1 - inside)`) so the detected object shows through undiminished. Each island renders as a soft-edged ellipse (feather from `m_detectionSoftEdgeDeg` = 3°) with a **dual modulation** after Veas et al. (2011): the object itself receives a mild saturation/brightness lift (`m_detectionEnhance` = 0.4) while a soft annulus just outside its boundary is slightly darkened (`m_detectionSurround` = 0.15), raising the object's centre–surround contrast from both sides without a glow artefact. The lift breathes gently at ∼1 Hz (`m_detectionPulseAmp` = 0.25), a low-frequency, low-amplitude temporal modulation chosen for attention capture at low annoyance (Waldner et al., 2014), far below the photosensitivity band. Detections expire after `m_detectionLifetime` = 0.6 s without renewal. The class filter is deliberately narrow: COCO classes 9 (traffic light) and 0 (person) only, the two object classes the Block B marker task uses as targets (Chapter 6).

4.8.2 The offline oracle pipeline

For evaluation, live inference is replaced by a **pre-baked detection track**, the oracle-perception instantiation of Chapter 6's methodology. A PC-side script (`Tools/bake_detections.py`) runs YOLO11 over the full-resolution 3840×2160 study video used in the test scene, as 3×2 overlapping tiles plus a full-frame pass merged by per-class non-maximum suppression, sampling at 0.5 s intervals, and writes a JSON track (`study_video.detections.json`) keyed by video time. The runtime loads it with `m_useBakedDetections = true` and plays detections back deterministically.

The bake over the 8-minute-39-second study clip produced **13,939 detections across 1,040 samples, with 99.5 % of samples containing at least one detection** (median 14 per sample, 2,952 traffic lights and 10,987 people). Unlike the live manager, the video test scene's manager (`VideoTestSceneManager`) draws on the shader's full sixteen-slot `_DetectionRects[16]` array (`k_maxDetections = 16`), which absorbs the oracle bake's density, so no ordering or survival policy is needed for the great majority of samples. These figures are the system's preliminary oracle-side data and reappear in Chapter 5 as one arm of the planned oracle-versus-live benchmark.

Oracle pipeline: source video → tiled YOLO11 full-res inference → NMS merge → JSON track → deterministic playback into shader islands.

Figure 4.9: The offline oracle pipeline, from source video to deterministic playback.

4.9 The Video Test Scene

All dynamic-scenario evaluation runs in a dedicated harness scene (`VideoTestScene.unity`, `VideoTestSceneManager.cs`) in which the passthrough layers are replaced by an 8-minute-39-second equirectangular driving clip decoded onto the sphere (`VideoPlayer` → RenderTexture → `_MainTexL`, with equirectangular sampling `_EqCamSampling = 1` and pole-safe blur tap folding). The scene preserves everything else, the same shader, window logic, modes, interaction, and detection islands, so a mode behaves identically over video and over the live camera feed, while gaining what live passthrough can never provide: **identical, repeatable, event-dense stimuli for every participant**, plus a deterministic ground-truth detection track. The harness is thus the simulation-based- evaluation instrument of the user study (Chapter 6, Block B) and the reason the study's stimuli are versionable artifacts rather than descriptions.

The harness carries its own alignment and perception plumbing. The equirectangular mapping exposes horizontal and vertical UV offsets (`m_videoUOffset`, `m_videoVOffset`) and a vertical flip (`m_flipVideoY`) so that any source clip's forward direction and horizon can be registered to the scene's coordinate frame without re-encoding the video. For live-inference experiments the manager maintains a dedicated 640×360 render target for the detector (`m_yoloRT`) so that YOLO never reads the 4K video texture directly. The downsample blit happens inside the detector component, on the detector's own cadence rather than every frame. For evaluation this path is bypassed entirely in favour of the baked track (`m_useBakedDetections = true`, the default). Mode robustness is handled at startup: any stale serialized mode from an older scene save is snapped to the current cycle's default (ColorPop), and the removed exploratory modes' shader toggles are zeroed once, since nothing drives them per-frame any more.

Two harness-specific notes have study consequences. First, in the video scene SignPop's grading covers the full sphere including the window (Section 4.5.2/4.5.5), which is the intended Block B configuration. Second, the video manager's motion-suppression master switch (`m_enableMotion`) currently defaults to **false**. Study Block B specifies suppression active, so the study configuration must set it explicitly, which the session sequencer asserts at start-up rather than leaving to the scene's saved state (Chapter 6, Section 6.10.1).

4.10 The Failure Museum

Documented dead ends are reusable knowledge. This section reports the failures that shaped the system, each with what was tried, why it failed, and what replaced it.

1. Screen-space camera sampling → double vision. The first shader sampled the camera texture by screen position. Per-eye the image was correct. Binocularly it could not fuse, because each eye sampled a different camera pixel for the same world point. On-device symptom: frank double vision. Replaced by world-direction sampling through a head-centred pinhole (Section 4.4), verified fused on device. Lesson: in stereo rendering, *eye-independence of the sampling domain* is a correctness criterion, not an optimisation.

2. Surface-projected passthrough → deprecated API. The obvious platform mechanism for "put real passthrough on my own geometry", surface-projected passthrough, was evaluated and rejected because it is deprecated as of SDK v83. Any architecture built on it would be built on removed ground. Replaced by the window-as-absence composite, which uses only stable compositor behaviour. Lesson: on young platforms, prefer primitives the OS must keep (alpha compositing) over conveniences it may withdraw.

3. `Shader.Find` on device → null. Runtime materials created via `Shader.Find("Unlit/Color"` worked in the editor and silently returned null on Quest: Android builds strip shaders that no serialized asset references, and the failure surfaced as a crashed startup coroutine rather than an error. Replaced by a serialized **template material** (`SelectionDotMat.mat`, assigned in the Inspector as `m_dotMaterialTemplate`) that runtime dots clone. The asset reference guarantees build inclusion. Lesson: never construct device materials from name lookups. Ship a serialized template.

4. Downsampled detection baking → missing traffic lights. The first oracle bake ran YOLO on a 640×360 downsample of the 4K source. Distant traffic lights fell below detectable pixel size and the track silently omitted most of them, an oracle that was not an oracle. Replaced by the full- resolution tiled PC-side baker (Section 4.8.2), raising coverage to 99.5 % of samples. Lesson: validate ground-truth pipelines against the *smallest* objects they must contain, and never let the convenience path (in-editor, low-res) overwrite the reference track (`BakeOnPlay` now defaults off).

5. Two operational traps, recorded for the protocol rather than the architecture. The platform's camera-based screenshot service returns black frames whenever the application holds the passthrough camera, because the two contend for the same resource, so the study's redundant session recording must use framebuffer capture (`screencap`/`screenrecord`) instead (Chapter 6). And the headset sometimes boots with controllers in `CONNECTED_INACTIVE`, leaving the application with no input at all and no way to distinguish that from a broken build, so the study build boots straight into the experiment scene, hand tracking is disabled, and waking the controllers is a line on the pre-session checklist. Lesson: input liveness is an environmental precondition, so it belongs in the protocol rather than the debugger.

4.11 Summary

One sphere, one shader, one manager: the implementation realises Chapter 3's design space with a per-fragment colour/alpha policy over a world-locked spherical window. Its load-bearing techniques, the window as absence, the hybrid composite of native passthrough and processed camera feed, and eye-independent world-direction sampling, are each answers to a specific platform wall, and its failures are documented with the same care as its successes because both are findings about building DR on consumer hardware. What the system *costs* (frames, milliseconds, heat) and how far its live perception stands from the oracle it is evaluated under are the questions of Chapter 5.

Chapter 5

Technical Evaluation: Plan and Preliminary Results

5.1 Purpose and evaluative stance

Chapter 3 argued that diminished reality (DR) on a consumer video-see-through (VST) headset is best understood as a design space organised by tiers of compositing access, and Chapter 4 described a reference implementation that occupies that space. Neither chapter, however, is entitled to its claims without measurement: "runs on consumer hardware", "native quality inside the window", and "the oracle assumption is a stated, bounded idealisation" are all empirical statements. This chapter specifies the benchmark protocol that will substantiate, or fail to substantiate, each of them (contribution C3).

Two properties of the protocol are worth stating up front, because they mirror the evaluative ethos of the user study in Chapter 6. First, every benchmark is **falsifiable**: each defines, in advance, the measurement method, the metric, and the criterion or interpretation band against which the result will be read. A benchmark that cannot fail is marketing, not evaluation. Second, the chapter is written **before the benchmarks have been run**. With one exception, the offline detection bake reported in Section 5.8, which was required during development, no measurement below has been performed. All protocol text is therefore in the future tense, and no result is anticipated. The completed benchmark tables will replace the placeholder rows in a later revision. The criteria will not move.

The benchmarks require no participants and are independent of the user study's own timeline, which is running ahead of them (Chapter 6, Section 6.11 reports its interim results). They will be executed against the same build version used for that collection, so that the numbers reported describe the exact binary participants experienced.

5.2 Apparatus and instrumentation

All measurements will be taken on the study hardware: a Meta Quest 3 (Snapdragon XR2 Gen 2, 8 GB RAM), running the Horizon OS build current at the study freeze date, with the study APK identified by build hash (Chapter 6, Section 6.10 pre-session checklist item 4). Instrumentation:

- **Perfetto system traces**, captured via the Meta developer tooling (`metavr` CLI / Perfetto capture), for GPU and CPU frame timing, render-stage breakdowns, and thread scheduling (Meta, 025a).
- **OVR Metrics Tool overlay and logcat telemetry** (Meta, 025b) for sustained frame rate, dropped/stale frame counts, battery level, and thermal warnings during long runs.
- **Through-the-lens photography** for the legibility benchmark: the headset mounted on a fixed stand, a digital camera with manual focus and exposure positioned at the left eyepiece, and printed optotype charts at marked distances under the study room's controlled lighting (Chapter 6, Section 6.3.3).
- **The offline detection oracle**: the full-resolution YOLO11 detection set produced by `Tools/bake_detections.py` over the study's driving footage (Section 5.8), used as ground truth for the oracle-gap benchmark.

Each benchmark will be run three times. Medians are reported with min–max ranges. Trace files, photographs, and analysis notebooks will be archived alongside the study data.

Reproducibility envelope. Every result in this chapter is specific to a tuple that will be stated once with the results and applies to all benchmarks: device serial, Horizon OS version, SDK version, APK build hash, refresh rate, eye-buffer resolution, and room lighting configuration. The platform's history within this project, a passthrough capability deprecated mid-development (Chapter 4), is the argument for recording the tuple explicitly: on consumer XR hardware, a benchmark without its platform coordinates is unreproducible within a single OS update cycle.

5.3 Benchmark T1: Per-mode GPU cost and frame-budget headroom

Claim defended. "The system runs within the display budget of consumer hardware", the precondition for every other claim in the dissertation.

Method. For each rendering configuration, a 120-second Perfetto trace will be captured in the study room with the headset worn (head gently moving, as tracking state affects load), and GPU frame time extracted per frame. Configurations:

Table 5.1: Rendering configurations benchmarked for GPU cost and frame-budget headroom (T1).

#	**Configuration**	**Tier (Ch. 3)**	**Scene**
1	Baseline: passthrough scene, all effects off	n/a	`CameraSphereVignette`
2	Soft Dark (overlay only, `_SimpleMode`)	2	`CameraSphereVignette`
3	Hard Dark (overlay only)	2	`CameraSphereVignette`
4	Blur (live PCA feed, two-ring 17-tap Gaussian + desaturation)	3	`CameraSphereVignette`
5	SignPop on video sphere (study Block B/C configuration)	3 (applied to video)	`VideoTestScene`
6	Baseline video sphere, effects off	n/a	`VideoTestScene`

Captures are taken worn rather than benched, because head motion exercises the code paths that matter, and CPU main-thread and render-thread times are extracted alongside GPU time, because an overlay sphere is cheap on the GPU while the manager scripts spend on the CPU.

Metrics. Median and 95th-percentile GPU frame time (ms), median CPU main and render thread time (ms), stale-frame rate (%), and headroom against the display budget (13.9 ms at 72 Hz, 11.1 ms at 90 Hz).

Criteria. *Pass* for a configuration used in the user study (2, 3, 5, 6): 95th-percentile GPU frame time within budget and stale-frame rate $< 1\%$, because jitter in a study stimulus is a confound, not merely a defect. Configuration 4 (Blur) is an *interpretation band* rather than a pass/fail: its cost relative to configuration 2 is the price of Tier-3 camera re-rendering against Tier-2 overlay compositing, which is a design-space datum in its own right.

5.4 Benchmark T2: Focus-window world-locking stability

Claim defended. "The focus window is world-locked": the rectangle stays fixed to the room, not to the head, under natural head motion. If the window visibly swims or lags, *Bar chart: median and p95 GPU frame time per configuration against the frame-budget line.*

Figure 5.1: Per-mode GPU frame time against the frame budget (T1).

the architectural claim of Chapter 4 fails, and the study's premise that participants attend to a stable spatial region weakens.

Method. Two complementary measurements. (a) *Log-level:* during scripted head sweeps (slow ∼30°/s and fast ∼90°/s yaw, three repetitions each), per-frame logs of head pose and the window's rendered azimuth/elevation bounds will be captured. Since the window is defined in world azimuth/elevation, its logged bounds should be constant while head pose varies. Any variation is implementation error. (b) *Display-level:* through-the-lens video at the camera's highest frame rate, with a printed reference edge aligned to the window edge. Frame-by-frame inspection gives the apparent displacement of the rendered edge against the physical edge during the sweep, in degrees of visual angle.

Metrics. Maximum apparent window-edge displacement (°) during the fast sweep; stale-frame rate during sweeps.

Criteria. *Pass:* apparent displacement $\leq 1°$ during the 90°/s sweep and no perceptible tearing in the through-lens footage. *Fail* triggers an implementation fix before the pilot's dry runs. This benchmark is deliberately scheduled before participant-facing sessions.

5.5 Benchmark T3: Legibility across rendering paths

Claim defended. The central architectural insight of Chapter 4: routing the focus window through the *absence* of overlay preserves native OS passthrough quality, while the Tier-3 camera re-render path is measurably worse. This asymmetry is the reason the architecture exists. It should be demonstrated, not asserted. Published psychophysics already shows Quest 3 passthrough falls short of natural vision acuity (Wang et al., 2026).

This benchmark measures the further gap *between the two in-headset paths.*

Method. A printed optotype chart (Landolt C or Snellen layout, sized to span logMAR-equivalent steps at the test distance) will be placed at 0.6 m (the Block A task distance) and photographed through the left lens under three display conditions with identical placement and lighting: (1) through the open focus window (native OS passthrough); (2) through the Tier-3 camera-feed path (the PCA sphere rendering the same region, effects set to identity); (3) direct photograph without the headset, as reference. Two additional captures at 2 m probe the far condition. Two independent raters (the author and one colleague, blind to condition labels on the crops) will score the smallest resolvable line per image.

Metrics. Smallest resolvable optotype line per path and distance, expressed as approximate logMAR steps relative to the no-headset reference; the *window-versus-periphery gap* in lines.

Method limitation, stated in advance. A camera at the eyepiece is not a human eye, so absolute acuity values from this method are indicative rather than clinical. The claim structure is therefore *relative*: same camera, chart, lighting, and placement across all three conditions, so the window-versus-periphery *gap* is meaningful even where the absolute values are not.

Criteria. *Interpretation, not pass/fail:* the architecture is vindicated if the window path resolves at least one optotype line finer than the camera-feed path at 0.6 m. If the two measure equal, the window-as-absence design loses its quality argument, though it retains the FOV and latency arguments, and the dissertation will say so.

Side-by-side through-lens crops of the same chart region: native window vs Tier-3 periphery vs no-headset reference.

Figure 5.2: Side-by-side through-lens crops of the same chart region.

5.6 Benchmark T4: The oracle gap, on-device detection versus the offline oracle

Claim defended. Part H's oracle-perception assumption (Chapter 6, Section 6.2) is legitimate only if it is stated as a *measured distance* from present reality, not a hand-wave. This benchmark produces that measurement, the single table that ties Part T to Part H.

One clarification keeps this benchmark honest. The user study uses no *live* detection anywhere. SignPop's colour-pop gate, its local window, and its saliency boost (Chapter 4, Section 4.5.5) are all driven by the offline oracle track baked in advance (Section 5.8). This benchmark therefore measures how far a *live* version of the exact mechanism the study tests would fall short of the conditions participants actually experienced, which is the size of what would have to close before the human-factors finding could be claimed for a deployed system.

Method. The deployable on-device configuration (YOLO11n, Jocher and Qiu, 2024, via Unity Sentis, Unity Technologies, 2024, FP16, at the camera-feed resolution the live pipeline would receive) will be run on the Quest 3 against the same study driving footage used for the offline bake, frame-matched to the oracle's 1,040 sampled frames. Oracle detections (full-resolution YOLO11 offline; Section 5.8) are treated as ground truth. A live detection matches an oracle detection when their boxes overlap at IoU ≥ 0.5 with the same class label.

Metrics and the oracle-gap table (template, values to be measured):

Table 5.2: The oracle-gap benchmark (T4): metrics comparing the offline detection oracle (assumption) against on-device live detection (reality), with the gap to be measured.

Metric	Offline oracle (assumption)	On-device live (reality)	Gap
Recall vs oracle (traffic lights + people)	1.00 by definition	*to be measured*	n/a
Precision vs oracle	1.00 by definition	*to be measured*	n/a
Throughput (inferences/s)	offline (not real-time)	*to be measured*	n/a
End-to-end latency, frame capture $\rightarrow$ detection available (ms)	0 by assumption	*to be measured*	n/a
Small-target recall (boxes $<$ 20 px at capture resolution)	1.00 by definition	*to be measured*	n/a

Matching procedure. Frame correspondence is exact, since both pipelines index the same video timestamps, so matching reduces to greedy per-frame box assignment by descending IoU, each oracle box matched at most once. Unmatched live boxes count as false positives and unmatched oracle boxes as misses. Live confidence thresholds will be swept and the operating point reported, so the headline recall is not an artefact of one threshold.

Threat to this benchmark, inherited by its consumers. The oracle is full-resolution YOLO11, not human annotation, so it is a *reference* whose own errors are unknown. Treating it as ground truth is defensible for measuring the gap between the offline and on-device pipelines, because both face the same footage and the offline one strictly dominates in resolution and compute. The absolute detection quality of either against human-verified truth is not established here, and no claim in this dissertation depends on it.

Criteria. *Interpretation bands, pre-stated:* live recall ≥ 0.8 of oracle at ≥ 10 inferences/s would place detector-driven guidance within near-term engineering reach, and the oracle assumption would be a modest idealisation. Recall below 0.5 or throughput below 2 inferences/s would mean the assumption is a genuine idealisation of present hardware, and RQ2/H2a/H2b's findings, which describe SignPop's effect under oracle-quality detection, would not yet be known to transfer to a deployed, live-detection system. Every detector-conditioned statement in Chapter 7 inherits that qualifier. The small-target row exists because distant traffic lights are precisely where a resolution-limited pipeline fails first (Section 5.8).

5.7 Benchmark T5: Thermal and battery endurance

Claim defended. The deployability argument of Chapter 3, Section 3.5: the camera-free Tier-2 dark modes, Soft Dark and Hard Dark, are claimed to be the only modes plausibly wearable for hour-long sessions. The prior expectation comes from the one published feasibility study of PCA-based on-device compositing, which projects 720p30 segmentation-based compositing for only 5–10 minutes before thermal throttling, a simulation-based estimate, not a measured on-device run (Laghari et al., 2025). The present system's Tier-3 path is shader-only, lighter than segmentation, so the outcome is genuinely uncertain, which is what makes the benchmark informative.

Method. Four 45-minute continuous runs on a fully charged, thermally rested device in the study room: (1) Soft Dark, passthrough scene; (2) Hard Dark, passthrough scene;

(3) Blur (live PCA feed), passthrough scene; (4) SignPop on the video sphere. Battery percentage, thermal-warning events (logcat), sustained frame rate, and stale-frame rate will be logged at one-minute resolution.

Pre-stated expectation (TB1). The camera-free configurations (1, 2, 4) complete 45 minutes without thermal throttling. The live-PCA configuration (3) may not. Confirmation would ground the dark-mode deployability claim empirically. Refutation in either direction is reported as found.

Study-feasibility criterion. Separately from TB1: the configurations the user study actually uses must sustain their block durations with margin, Hard Dark for ≥ 18 minutes (Block A plus tutorial) and the video-sphere configuration for ≥ 19 minutes (Block B). Failure here would require a protocol change and would be caught before the pilot's dry runs. Note that the study, by design, never runs live PCA: the longest-exposure mode (Hard Dark) is a Tier-2 overlay, which is what makes the 48.5-minute session thermally plausible in the first place.

5.8 Preliminary results: the offline detection bake

One component of the technical programme has already been executed, because development required it: the offline detection oracle over the study's driving footage.

Why full resolution. Detections that serve as ground truth or as pre-scripted salience input must be computed at a resolution the live pipeline cannot afford, which is exactly what makes them an *oracle* in the sense of Section 5.6. Distant traffic lights are the binding case. They are fully visible to a human viewer but subtend few pixels, so they fall below detectable size in a downsampled frame.

Procedure. `Tools/bake_detections.py` ran YOLO11 at full frame resolution over 1,040 samples at 0.5 s intervals, spanning the 8-minute-39-second study clip (`study_video.mp4`), detecting traffic lights and people.

Results. The bake produced **13,939 detections across the 1,040 sampled frames**, with **99.5% of sampled frames containing at least one detection** (median 14 per sample, 2,952 traffic lights and 10,987 people). Two properties of this result matter for the rest of the dissertation. First, it confirms the footage is genuinely event-dense, a precondition for its role as the Block B stimulus. Second, it establishes the oracle side of the T4 table: a fixed, versioned, full-resolution detection set against which the live pipeline can be scored, rather than a moving target.

What this result does not establish. It says nothing about on-device feasibility (T4's live column), nothing about detection quality in absolute terms (YOLO11 at full resolution is the *reference*, not verified truth against human annotation, a limitation inherited by T4 and stated there), and nothing about the user-facing system, which uses no detections at all in the study configuration.

5.9 Reporting rules

Three rules bind the results revision of this chapter. Results will be reported for every benchmark attempted, including failures and anomalies, with trace archives retained. No criterion or band stated above will be adjusted after the data are seen. And where a benchmark is dropped for time, the pre-agreed sacrifice order putting T5 first, the drop is stated rather than silently omitted.

Chapter 6

User Study Design

Status note. This study targets N = 20 analysed participants, run to a frozen protocol after a formal pass/fail pilot. That collection is **still running** at the time of writing, not finished. Fifteen people have been tested across an initial pilot day and four further collection days, preceded by informal, naive-pilot iteration used to tune stimulus parameters and task form ahead of the frozen protocol. Fourteen usable pairs exist for Block A and thirteen for Block B. Section 6.11 reports the interim results from those pairs, explicitly as interim: recruitment continues toward the N = 20 target, and nothing in Section 6.11 is the pre-registered confirmatory analysis this chapter specifies.

6.1 Rationale and Methodological Positioning

6.1.1 What the study must be able to conclude

The purpose of this study is to determine whether the diminished-reality (DR) attention-guidance system described in Chapters 3 and 4 *works*, and, just as importantly, to be able to determine that it *does not* work, if that is what the data show. This dual requirement shapes every design decision in the chapter. A study that can only confirm is not an experiment. It is a demonstration. Two design choices in particular produce that defect: an underpowered between-subjects comparison, and safety hypotheses phrased as bare nulls ("peripheral detection holds steady", "accuracy is maintained or improves"). A non-significant difference in a small sample is evidence of nothing, so a design resting on bare nulls has no way to fail informatively.

The present design removes the possibility of an inconclusive outcome by reframing the research question around an effect that is already established in the literature. Peripheral visual distractors reliably impose a measurable performance cost on a focal sustained-attention task in a head-mounted display: in a virtual-classroom Go/No-go study with 66 participants, commission errors more than doubled in the presence of peripheral distractor events ($1.33 \rightarrow 3.15$, $p < .001$), with omission errors showing the same pattern (Ai et al., 2025). The primary question is therefore not the unconstrained "does the vignette make people better?" but the falsifiable:

> *Peripheral distractors impose a measurable cost on a focal task. Does the DR vignette reduce that cost?*

Under this framing, every outcome is informative. If the vignette works, the error rate under Filter is lower than under NoFilter. If the vignette does nothing, the two error rates are indistinguishable, a bounded, conclusive negative. The design relies on the established literature cited above (Ai et al., 2025, N = 66) for confidence that the distractor reel imposes a real cost to begin with, rather than on an internal manipulation check. There is no configuration of results that yields "inconclusive".

6.1.2 Simulation-based evaluation and the oracle-perception assumption

The study occupies a deliberate position between concept evaluation and artefact evaluation, mirroring the dissertation's two-part structure (Chapter 1). Two named methodological commitments govern this position.

First, the dynamic (driving) block is a **simulation-based evaluation**. Participants view pre-recorded equirectangular driving footage rendered on the system's sphere rather than a live street scene, in the tradition established by Cheng et al. (2022), who evaluated DR concepts entirely inside VR simulations because live DR was not feasible. Simulation gives every participant identical, repeatable, event-dense stimuli, something no real environment provides, at a known cost to ecological validity that is declared rather than obscured: no claim in this dissertation concerns driving, drivers, or road safety. The driving footage is a dynamic, high-event-rate monitoring stimulus, nothing more (see the claim-scoping rules in Chapter 1).

Second, the perception problem is **decoupled from the attention question, but from *live, on-device* detection specifically, not from detection itself**. SignPop is not detector-free. Its colour pop, its detection aperture, and its saliency lift are all gated on one detection signal (Chapter 4, Section 4.5.5), and that signal comes from an oracle, the offline full-resolution bake, computed once in advance rather than by a model running during a session. This lets the study ask whether an effect built on excellent detection helps attention, without first building excellent detection on-device. The finding therefore does not depend on today's on-device detection quality, but it does depend on oracle-quality coverage, and Chapter 5's oracle-versus-live benchmark measures that gap. Nor does the design distinguish which of the gated effects is responsible for any measured benefit, which Section 6.10.2 carries forward as a stated limitation.

Critically, the two confirmatory blocks sit at opposite ends of this spectrum. Block A (workstation) runs on **real passthrough with real physical distractors**. It evaluates the actual artefact as built. Block B (driving) runs on the simulated video scene. It evaluates the concept under the ideal-context assumption. Convergence between them strengthens both. Divergence is itself a reportable finding about what simulation-based XR evaluation misses.

6.1.3 Why within-subjects with repeated trials

The design is fully within-subjects, with no between-subjects factors, targeting **N = 20 analysed participants** and a high trial count per condition. Three considerations justify this choice over the superficially more ambitious alternative of a larger between-groups design.

First, precedent: the two closest published DR evaluations are within-subject, at N = 16 and N = 12 (Cheng et al., 2022) and as two mixed-model experiments (McLaughlin et al., 2025). Second, field norms: Caine (2016) found the modal CHI sample size to be 12, with in-person studies drawing their power from repeated measures rather than headcount. Third, arithmetic: each participant contributes 140 scored task events per Block A condition and 40 marker events per Block B condition, so twenty participants yield on the order of 5,600 Block A trials, which trial-level mixed models can exploit fully while per-participant aggregates feed classical paired tests as robustness checks. A between-subjects design at feasible recruitment numbers would discard exactly the within-person variance control that makes small-N studies conclusive.

6.2 Research Questions and Pre-Registered Hypotheses

RQ1 (primary). Does peripheral DR dimming (Hard Dark, world-locked window) reduce the performance cost that peripheral visual distractors impose on a focal workstation task?

RQ2 (secondary). Does SignPop's detection-gated salience re-grading improve detection of gated objects specifically where the gate has room to act? That is the band that is neither so close to the focus centre that the object is already fully visible, nor so far into the periphery that it falls outside useful glance range. And does it do so *without* degrading detection pooled across the whole field of view beyond an acceptable margin?

RQ3 (exploratory). After experiencing the system, what do participants report about perceived focus benefit, perceived awareness cost, and visual comfort?

Every hypothesis below is stated with the observation that would refute it. A hypothesis without a refuting observation is not admitted to the design.

Table 6.1: Pre-registered research hypotheses, their refuting observations, and confirmatory status.

ID	Hypothesis (directional, falsifiable)	Refuting observation	Status
H1	Error rate on the Block A task, run with distractors present throughout, is lower with the vignette On than Off, by at least the smallest effect size of interest (SESOI; default $\geq$ 50% reduction from the vignette-off error rate, finalised from pilot data).	The paired difference's 90% CI excludes the SESOI (benefit too small to matter), or the difference is absent or reversed.	Primary confirmatory
H2a	With SignPop On, marker-detection hit rate in the 20–30° eccentricity band is higher than with the vignette Off. This is the band where the colour-pop gate has visible room to act but the target is still within useful glance range (Section 6.6.3).	Band-level difference's CI includes 0 or favours Off.	Secondary confirmatory
H2b	With SignPop On, marker-detection hit rate pooled across the whole field of view is *equivalent* to Off within a margin of $\Delta = 10$ percentage points (two one-sided tests, TOST).	Equivalence not established and the point estimate shows a drop $>$ 10 pp. A conclusive finding that the mode harms situational awareness.	Secondary confirmatory (safety)

Table 6.1: Pre-registered research hypotheses, their refuting observations, and confirmatory status.

ID	Hypothesis (directional, falsifiable)	Refuting observation	Status
EQ1–EQ3 (exploratory)	End-of-session questionnaire responses favour the effects on perceived focus benefit (EQ1), perceived awareness cost (EQ2), and visual comfort (EQ3).	n/a (exploratory, no confirmatory claim attaches)	Exploratory

SESOI rationale. H1's smallest effect of interest is expressed as a *proportion of the vignette-off error rate* rather than as a raw error count, because the raw error rate depends on stimulus tuning that the pilot legitimately adjusts. The 50% default embodies a defensible practical judgement: a vignette that removes less than half of the distraction-laden error rate does not justify wearing a headset to obtain it. The pilot converts this proportion into an absolute error-rate margin for the power check before the main study commits, and this conversion is recorded in the post-pilot lock-in memo (Appendix A). H2b's 10-percentage-point margin reflects the safety framing inherited from the DR focus-versus-awareness trade-off documented by McLaughlin et al. (2025) and Murph et al. (2021): in a monitoring context, losing more than one detection event in ten to the overlay, pooled across the whole scene, is an unacceptable awareness cost. Equivalence testing follows Lakens (2017), which gives the safety hypothesis a form that can fail rather than a bare null.

6.3 Participants, Ethics, and Apparatus

6.3.1 Participants

Twenty-four adults will be recruited (target: 20 analysed, with four spares against dropouts and pre-registered exclusions). Inclusion criteria follow the approved ethics protocol: aged 18 or over, normal or corrected-to-normal vision, no self-reported vestibular disorders. Prior VR experience is recorded on a four-level scale (never / a few times / monthly / weekly or more) and used as a robustness covariate rather than a selection criterion, since the deployed system must serve VR-naïve users.

6.3.2 Ethics and safety

The study runs under the ethics approval obtained for the protocol, with every difference between the approved protocol and the design as run classified and reviewed with the supervisor beforehand. The substantive task classes, seated video viewing, button presses, and tablets as distractors, are unchanged from what was approved.

Sickness risk is managed by design and by rule. By design, all tasks are seated, there is no artificial locomotion, total headset time is approximately 35 minutes, and a headset-off break separates the two confirmatory blocks. By rule, a written **stop rule** is applied without discussion: the VR portion terminates immediately if the participant reports nausea or dizziness at a moderate or worse level, at a scheduled comfort check or spontaneously, or if the experimenter observes distress. Comfort checks fall at the break after Block A and mandatorily after Block B before headset removal, with additional checks on request. After termination the participant rests seated, is offered water, and remains until symptoms subside. Data collected up to that point are retained per the consent wording and the participant is replaced from the spare pool. The Virtual Reality Sickness Questionnaire (VRSQ; Kim et al., 2018) is administered **pre and post** session, because post-only sickness scores are uninterpretable without a baseline (Brown et al., 2022).

6.3.3 Apparatus

All sessions use the same Meta Quest 3 headset with controllers (hand tracking disabled), in the same room, with bright constant lighting at a marked setting. Quest 3 passthrough acuity degrades markedly in low light (Wang et al., 2026), so lighting is a controlled variable, not a convenience. The participant sits on a fixed chair at a desk. The experimenter monitors the participant's view via an `scrcpy` mirror with screen recording running as a redundant data channel. Every stimulus in every block is deterministic and scripted from versioned schedule files. No live object detection runs anywhere in the study. The instrumentation, a telemetry component, a condition sequencer, a marker scheduler, and a post-session validator, is described in Section 6.10.1.

6.4 Design Overview

6.4.1 Structure

Table 6.2: Overview of the within-subjects design: participants, session length, blocks, and counterbalancing.

Property	Value
Design	Fully within-subjects, no between-subjects factors
N	20 analysed (24 recruited)
Session length	~48.5 minutes including consent and debrief
Blocks	A, workstation, Filter vs NoFilter (primary); B, driving marker detection, Filter vs NoFilter (secondary); C, end-of-session questionnaire, headset off (exploratory)
Counterbalancing	Block A: condition order alternated by participant parity (Filter-first / NoFilter-first). Block B: condition order alternated by participant parity (10 SignPop-first, 10 Off-first), with the two marker sets alternated against condition. Block C carries no conditions to counterbalance.
Block order	Fixed A→B→C for all participants: A is primary and must come freshest. C is a headset-off questionnaire and comes last. Acknowledged as a limitation (Section 6.10.2).

Within each block, the stimulus material, seating, lighting, controller, and task are identical across conditions. Only the shader state differs. In vignette-off conditions the focus window is still painted and locked by the participant, so the motor and procedural experience is identical across conditions. The vignette is isolated as the sole independent variable.

6.4.2 Why these modes, and why these pairings

Ten peripheral treatments exist in the codebase (Chapter 4). Two of them, Hard Dark and SignPop, were carried to confirmatory testing in Blocks A and B. This section explains that split before it explains why each tested mode was paired with its block.

The tested modes have data behind them. Hard Dark and SignPop were run on real participants, and Chapters 5 to 7 report what happened. Every claim this dissertation makes about them is a claim about measured behaviour, not a design argument.

The other eight modes were set aside for one of four reasons, not one. First, two are ruled out by prior literature. Blur was the most noticed and most disliked of five peripheral techniques in its own source study, actively irritating 59 of 102 participants (Grogorick et al., 2018), and it also carries the heaviest shader cost of any mode (Chapter 4). ChromaticCool loses to luminance-only modulation in its own source study (Bailey et al., 2009). Testing either again here would repeat a result the field already has. Second, informal piloting narrowed a larger candidate set down to the two modes early testers found most comfortable and effective, and four modes lost that comparison for reasons specific to each. ConspicuitySqueeze brightened the periphery instead of squeezing its contrast toward the local mean, the opposite of what it was meant to do, confirmed by inspecting the shader directly. OutlinedDark's surviving edges read as a comic-book style that testers found more interesting than the task itself. GranulatedPeriphery's grains occlude fine detail, which is exactly the wrong failure mode for a driving scene where the detail is the target. SpotLift could not even be built on the passthrough path: it would brighten the focus window rather than darken the rest, and the OS passthrough layer cannot be brightened, so it exists only in the video path. All four are reported in Chapter 4 as design points, not as tested claims. Third, ColorPop is not itself set aside so much as superseded: SignPop is built directly on ColorPop's grading pipeline with a detection gate added, so ColorPop was never a candidate for independent testing alongside its own extension. Fourth, one mode's expected effect was judged too small to detect with the session budget and sample size this study can afford: attenuating the periphery to 75% opacity is a smaller, harder-to-power claim than removing it completely. Soft Dark is therefore not tested, and this dissertation reports no participant data on it of any kind (Section 6.7).

Hard Dark is assigned to Block A because dark modes *remove* peripheral signal, and the workstation paradigm measures precisely what removal should buy: a reduced distractor cost. Hard Dark is preferred for the confirmatory test because it is the strongest manipulation in the family, maximising the probability of detecting the mechanism if it exists.

SignPop is assigned to Block B because it does not primarily remove signal. It *re-grades salience*, gated on a detected object rather than colour class alone. The marker-detection paradigm measures what re-grading should buy, a detection benefit concentrated where the gate fires (H2a), and what it must not cost, peripheral awareness overall (H2b).

The mode × scenario confound is stated plainly. Hard Dark is tested only at the workstation and SignPop only in the driving scene. All confirmatory claims in this dissertation are therefore *mode-in-context* claims ("Hard Dark reduces distractor cost in a workstation task"), never mode-generalised claims ("dark vignettes work"). Nothing in the study softens this confound, because no participant ever saw two modes on a common stimulus.

6.5 Block A: Primary, Workstation Focus (Hard Dark, real passthrough)

Block A answers the primary question: does blacking out the periphery around a world-locked focus window reduce the errors that real peripheral distractors impose on a focal task?

6.5.1 Design

The comparison is **Filter on vs Filter off**, with the distractor reel present throughout, run as a direct paired comparison of error rate. H1 is evaluated on this comparison.

6.5.2 Physical layout and distractor stimulus

The participant sits at a desk with the task surface directly ahead: a world-locked virtual panel (Section 6.5.4) approximately 0.6 m from the eyes. Two tablets on stands are placed at $\pm$**35° azimuth** from the task centre, approximately 1 m from the participant, screens facing them. The 35° eccentricity places the distractors in the near-periphery zone of maximal involuntary attentional capture identified by the Useful Field of View literature (Ball and Owsley, 1993). Note that Ball & Owsley define and validate the UFOV *test*; the $\approx$30° extent used here is the conventional figure from that literature rather than a value stated in their paper.

All fifteen people tested (Section 6.11) were run in-headset on this apparatus. Task feel and timing were tuned beforehand on a browser prototype (this chapter's status note), which placed its distractors at only 15–25° of eccentricity and so was used for timing alone. Hard Dark's job in this block is to occlude the tablets from view entirely, not merely dim them at a distance.

The tablets play a prepared **distractor reel**: fast-cut, high-motion, bright video with scheduled salient event bursts, hard cuts to full-screen colour flashes with sudden motion, at fixed timestamps approximately every 20–30 seconds, identical across participants. It runs continuously, in every condition, for every participant.

6.5.3 Conditions

The two conditions are **NoFilter** (`_DrIntensity = 0`, so the window is painted but invisible) and **Filter** (Hard Dark on, window live). The distractor reel plays on both tablets throughout each.

plan-view diagram of the Block A layout: participant, desk, task surface, tablets at ±35°, focus-window extent.

Figure 6.1: Plan view of the Block A apparatus and distractor placement.

Each condition lasts 4 minutes 40 seconds, which is 140 trials at the 2.0 s stimulus onset asynchrony fixed in Section 6.5.4. The two conditions, Filter and NoFilter, run once each per participant, in counterbalanced order. Before the first condition the participant paints the focus window around the task surface with the right trigger, the system's normal interaction, and releases to lock it. The experimenter verifies that both tablets fall outside the window and the entire task surface falls inside it. The same locked window persists across both conditions, re-verified between them. Only `_DrIntensity` toggles.

6.5.4 Task

The task form answers a genuinely optical rather than attentional risk: Quest 3 passthrough does not reach normal visual acuity, and a task participants cannot comfortably *see* would floor out in every condition for reasons unrelated to the vignette. The task is therefore rendered on a virtual, world-locked panel rather than read through passthrough. Gate G1 (Appendix A) confirms it performs in range.

The task, forced-choice 1-back. A world-locked panel inside the focus window presents one look-alike shape at a time (drawn from a set of six, $\geq 3°$ visual angle), and the participant answers YES/NO on every shape, "same as the previous shape?", via trigger press. This is a forced-choice design rather than Go/No-go, because a Go/No-go stream is overwhelmingly targets and ceilings out (96–100% accuracy in piloting) regardless of distractor condition, which would leave the block unable to detect any vignette effect at all. The forced-choice 1-back adds a working-memory and goal-maintenance component on top of the reactive one, and its memoryless 50%-repeat sequence keeps performance off ceiling. Timing is frozen for the study at 140 trials per condition, a 2.0 s stimulus onset asynchrony, and a 1.7 s answer window. The first shape of each run is a memoriseonly trial with no scored response. Outcomes are `hit` (YES on a genuine repeat), `miss` (a repeat missed, by wrong answer or timeout), `commission` (YES on a non-repeat), `correct_reject` (NO on a non-repeat), and `no_response` (timeout on a non-repeat, logged separately as an engagement signal). Every onset and response is logged to the millisecond. Error rate is misses plus commissions, and reaction time is taken from correct trials. No per-trial correctness feedback is given during scored runs, only pacing feedback ("too slow"), so trial-by-trial feedback cannot shift participants' speed/accuracy strategy mid-run.

Implementation. The task runs in the in-headset panel, ported from a standalone browser prototype (`PilotTools/block-a-cpt.html`) used for no-headset timing work. The window itself uses room-geometry anchoring for this block specifically, not the bearing-and-elevation direction-lock that is the system's default and that Block B uses (Chapter 4). Four corner rays are captured when the participant releases the paint trigger and remembered as physical points in the room, so the window stays put as the participant leans toward or away from the desk.

6.5.5 Measures

Table 6.3: Block A measures and their instruments or ground truth.

Measure	**Instrument / ground truth**
Error rate per condition (primary DV input)	StudyLogger event log
H1 statistic = errors(NoFilter) − errors(Filter)	Derived per participant. Also modelled at trial level
Reaction time	StudyLogger, milliseconds
Head-turns toward tablets ($> 30°$ yaw from task centre): count and dwell	Head-pose telemetry, 60 Hz
Noticing check	One open question after Block A: "Did anything make the side screens easier or harder to ignore?"

6.6 Block B: Secondary, Driving Marker Detection (SignPop, video scene)

Block B answers the secondary pair: does SignPop improve detection specifically where its detection-gated re-grading has room to act (H2a), without degrading detection pooled across the whole scene beyond the safety margin (H2b)?

6.6.1 Design

The comparison is **Filter on vs Filter off**, and the task uses the driving footage's own content as the target. A ring marker is authored onto each real traffic-light or pedestrian ("person") object as it appears in the footage, and the participant presses the trigger on noticing it. Putting the target on a real object, rather than on an abstract probe overlaid on the scene, is what makes the measurement direct. The object under the marker is exactly the kind of object SignPop's detection gate is built to reveal, so the task measures detection of the thing the filter actually acts on. H2a and H2b are both evaluated on this comparison.

6.6.2 Stimulus and conditions

The block uses the system's video test scene: equirectangular urban driving footage rendered on the display sphere. Both conditions play the *same* clip, `study_video.mp4`, in full, running 8 minutes 39 seconds each. Holding the footage fixed means the two conditions differ only in shader state and in which marker set is scored, never in what the participant is looking at. The focus window is pre-set to the windscreen region, direction-locked in azimuth/elevation for all participants (Chapter 4). There is no window painting in this block, so the guided region stays constant across the sample.

Table 6.4: Block B conditions: shader state for the driving marker-detection task.

Condition	**Shader state**
NoFilter	Vignette Off (raw video)
Filter	SignPop On (motion suppression active, shipping defaults, detection gate enabled)

Condition order alternates by participant parity, and the two marker sets are paired with the two conditions in alternation as well, so neither set is systematically tied to Filter or NoFilter.

6.6.3 Marker-detection task

Forty scored markers per condition, drawn from two disjoint sets of forty over the same clip. That is 880 marker presentations in total across the eleven usable pairs collected so far. Using a different set in each condition stops a participant meeting the same target twice, which would turn the second condition into a memory test. Each marker is a ring authored onto a real traffic-light or pedestrian target in the footage. It fades in rather than switching on, because sudden onsets capture attention automatically and would swamp the effect being measured (Yantis and Jonides, 1984). It also modulates the pixels underneath rather than covering them, so it keeps its own internal contrast even where the filter has dimmed everything under it (Wallis et al., 2015). The marker itself is dimmed by the same falloff as the rest of the scene regardless of condition, so a filtered marker is never easier to see as a marker. Only the object beneath a detected marker is treated differently between conditions. Seventy of the eighty authored targets (88%) fall inside an active detection window while on screen, replicating the runtime's own gates: minimum lifetime 0.6 s, minimum box size 1.2°, and a 16-window cap. In the NoFilter condition the detector is suppressed along with the rest of the effect, so the same object receives no pass-through there.

A **hit** is a trigger press while the marker is on screen, within 10° of its position at that instant. Presses that land on nothing are **false alarms**. Markers move while visible, by 17° of travel on average, so scoring against a marker's *average* position would misassign a meaningful share of presses to the wrong analysis band. Instead every press is scored against the marker's reconstructed position, from the recorded scene track, at the exact instant of the press. Practice markers, which use ids in a reserved range and run before scored trials, are excluded from every analysis.

Eccentricity bands. For the secondary, exploratory analysis behind H2a, each scored marker's position at the instant of its catch (or, for misses, its mean on-screen position) is converted to an eccentricity from the focus-window centre. It is then assigned to one of four bands, fixed from published gaze landmarks rather than chosen after seeing the data: under 10°, the road-centre region where driver gaze concentrates (Victor et al., 2005); 10–20° and 20–30°, spanning the published probe range for spare-capacity measurement (Jahn et al., 2005; ISO, 2016); and beyond 30°, past the conventional useful-field-of-view extent (Ball and Owsley, 1993). H2b's safety check pools hit rate across all four bands, that is, across the whole field of view.

6.6.4 Measures

Table 6.5: Block B measures and their instruments.

Measure	Instrument
Hit rate pooled across the whole field of view (H2b input)	Authored-marker click log

Table 6.5: Block B measures and their instruments.

Measure	Instrument
Hit rate by eccentricity band, esp. 20–30° (H2a input)	Authored-marker click log + reconstructed scene track
Reaction time on hits	Authored-marker click log
False alarms per condition	Authored-marker click log
Head yaw/pitch telemetry and angular speed	StudyLogger, 60 Hz

Head pose serves as the attention proxy in place of eye tracking, which the Quest 3 does not provide. Head orientation is a validated gaze surrogate at the eccentricities used here (Higgins et al., 2022; Sitzmann et al., 2018).

6.7 Block C: Exploratory, End-of-Session Questionnaire

Block C answers RQ3: after experiencing the system, what do participants report about it? It carries no confirmatory claims.

Protocol deviation. A three-mode sampler was specified here, 75 seconds each of SignPop, Soft Dark, and Hard Dark on common footage with Likert items after each. It was cut from the running protocol, because the two confirmatory blocks already account for roughly 35 minutes in the headset and the sampler would have added two further exposures at the point of maximum fatigue. Two consequences follow, and both are carried through this dissertation rather than absorbed quietly. Soft Dark was never shown to a participant, so no participant data on it exists. And no participant ever saw two modes on a common stimulus, so the mode × scenario confound of Section 6.4.2 is unmitigated.

The instrument. A single end-of-session questionnaire (ESQ), self-completed with the headset off, after the post-session VRSQ and before the study-aims debrief. Opinions are captured before the hypotheses are revealed, which is the demand-characteristics defence of Section 6.10.2. It takes about five minutes: 23 rated items, one ranking, one multi-select, and four open questions. The rated items are grouped into three constructs, each mixing positively and negatively keyed items to counter acquiescence bias:

1. **Perceived focus benefit** (EQ1): did the effects subjectively help concentration on the task area?

2. **Perceived awareness cost** (EQ2): did the effects make the participant feel cut off from, or late to notice, what was around them?

3. **Visual comfort** (EQ3): were the effects comfortable to look at over a session, and did the window boundary itself intrude?

The open questions ask what the participant would change, what would stop them using it for an hour of real work, and what the edges of their vision felt like.

Caveats, stated honestly. The questionnaire asks for one considered opinion of the whole session, so it cannot separate Hard Dark's contribution from SignPop's. It is a system-level instrument, reported as exploratory, with medians and the open answers quoted rather than tested.

6.8 Session Procedure and Pilot

6.8.1 Session timeline

The full session runs ~48.5 minutes, inside the 40–60-minute tolerance window participants typically accept. Headset-on time is ~35 minutes of that. The information sheet is sent in advance, and the closing questionnaires are done with the headset off.

Table 6.6: Session timeline: clock time, duration, and activity for the full ~48.5-minute session.

Clock	Duration	Activity
00:00	4 min	Welcome; written consent (information sheet pre-read); demographics + VR-experience form; VRSQ (pre)
04:00	4.5 min	Headset fitting and comfort check; tutorial: paint/lock a practice window; practice to criterion ($\geq$ 90% accuracy on a 1-minute practice stream; marker practice: 6 markers, $\geq$ 4 hits). One repeat allowed.
08:30	1 min	**Block A**: paint and lock focus window; experimenter verifies geometry
09:30	4 min 40 s	Condition i (Filter or NoFilter, order counterbalanced)
14:10	1 min	Reset (tablets re-cued, window geometry re-verified)
15:10	4 min 40 s	Condition ii
19:50	0.5 min	Noticing question
20:20	1.5 min	Headset-off break; verbal comfort check (stop rule applies throughout); water offered
21:50	8 min 39 s	**Block B**: condition i (Filter or NoFilter, order counterbalanced), the study clip in full

Table 6.6: Session timeline: clock time, duration, and activity for the full ∼48.5-minute session.

Clock	**Duration**	**Activity**
30:30	0.5 min	Reset
31:00	8 min 39 s	Condition ii, the same clip in full with the other marker set
39:39	0.5 min	Fatigue and comfort check; headset off
40:09	1 min	VRSQ (post)
41:09	5 min	**Block C**: end-of-session questionnaire (ESQ), self-completed
46:09	2.5 min	Study-aims debrief; thanks
48:39	n/a	Experimenter runs the session validator **before the participant leaves**. Incident log entry if anything deviated

Arithmetic: opening 8.5; Block A 11.8 (1 + 2×4.67 + 1 + 0.5); break 1.5; Block B 18.3 (2×8.66 + 2×0.5); close 8.5 (1 VRSQ + 5 ESQ + 2.5 debrief). Total 48.6 minutes.

The practice-to-criterion tutorial serves a specific validity function beyond familiarisation: the practice includes the vignette itself, so its first appearance is never during a measured trial, neutralising first-exposure novelty effects at the point where they would otherwise contaminate data.

6.8.2 Pilot gates

A pilot of three to five participants, excluded from the main sample, ran before collection opened. Its purpose was structural rather than exploratory: to make an uninterpretable main study impossible. Three gates had to pass before launch, covering task form, marker calibration, and two full end-to-end protocol dry runs, each with a stated pass criterion and a stated action on failure. Stimulus parameters could be tuned between pilot participants. Hypotheses, margins, and analysis choices could not. The gates are reproduced in full in Appendix A, together with the pre-registered exclusion and data-quality rules and the verbatim participant scripts. The pilot concluded with a one-page lock-in memo recording the chosen task form, the tuned parameters, and the finalised SESOI for H1, after which nothing changed until collection ended.

6.9 Analysis Plan and Decision Table

All analysis will be performed in Python (pandas, pingouin, statsmodels) or R, with scripts written and dry-run on pilot data before main collection begins. $\alpha = .05$ throughout. Effect sizes and confidence intervals are reported for every test: 95% two-sided by default, 90% where one-sided or equivalence logic applies (the H1 SESOI bound and the H2b TOST).

6.9.1 Confirmatory analyses

Table 6.7: Confirmatory analysis plan: primary test, robustness check, and effect size per hypothesis.

Hypothesis	Primary test	Robustness check	Effect size
H1	Linear mixed model on trial-level errors (logistic LMM: error $\sim$ vignette + (1 | participant)); the vignette term is the test	Paired t-test on per-participant error rate, NoFilter vs Filter; Wilcoxon signed-rank if Shapiro–Wilk rejects normality	dz on the paired difference; odds ratio from the LMM
H2a	Paired t-test (Wilcoxon fallback) on per-participant hit rate in the 20–30° eccentricity band, Filter vs NoFilter	LMM on trial-level hit/miss within the band (logistic)	dz
H2b	TOST on hit rate pooled across the whole field of view (paired), margin $\Delta = 10$ pp (Lakens, 2017)	Bayesian paired test as an optional sensitivity check	Mean difference with 90% CI against the margin

Mixed-effects modelling at trial level follows the analysis approach of McLaughlin et al. (2025). The aligned-rank-transform alternative for nonparametric factorial data (Wobbrock et al., 2011) is noted but not preferred, in light of recent critiques of ART's error control (Tsandilas and Casiez, 2024).

Multiplicity policy. H1 is the single primary endpoint and receives no correction. H2a and H2b form the secondary family, Holm-corrected across the two. Everything else, the end-of-session questionnaire and the head-turn telemetry, is labelled exploratory and is never quoted as a confirmatory finding.

Power. For the primary contrast, computed per participant as a paired difference in error rate between NoFilter and Filter, a paired t-test at $\alpha = .05$ (two-tailed) with N = 20 achieves 80% power for dz $\approx$ 0.65. The distractor-driven error rate underlying this comparison is large in the source literature (commission errors more than doubling at N = 66), so a vignette recovering half of it is plausibly detectable at this N. The pilot verifies the locally achievable error rate before the main study commits, and if it proves small, the SESOI conversation with the supervisor happens *before* data collection rather than after. Trial-level mixed models over ~5,600 Block A trials provide additional sensitivity beyond the aggregate-level power figure.

6.9.2 Exploratory analyses

End-of-session questionnaire items are reported per construct as medians with interquartile ranges, with the open answers quoted rather than coded, since a single end-of-session opinion per participant supports description and not testing. Head-turn counts and dwell toward the tablets (Block A) and head angular speed (Block B) are analysed descriptively with paired tests, flagged exploratory. Interview responses are thematically coded into a compact table (mode → recurring likes and concerns), with quotes used to illustrate the quantitative findings. H1 is re-estimated within VR-experienced and VR-naïve subgroups as a robustness description, not a test.

6.9.3 Decision table

The following table is pre-committed, and the dissertation will report whichever row occurred once the $N = 20$ collection concludes. The interim results of Section 6.11 are consistent with row 1: H1 is supported, and H2b's refuting observation did not occur, so no harm to situational awareness is shown. The formal two-sided equivalence statistic is not established at the interim N, for the one-directional reason set out in Section 6.11.2. These are not this table's confirmatory trigger. The table maps outcomes of the final, fully powered collection, not an interim analysis run partway through recruitment.

Table 6.8: Pre-committed decision table mapping outcome patterns to dissertation conclusions.

#	**Outcome pattern**	**Dissertation conclusion**
1	H1 supported (paired difference in predicted direction, $\geq$ SESOI) **and** H2b passes equivalence	**The system works**: peripheral DR dimming reduces distraction cost, and the salience mode does not measurably harm peripheral awareness.

Table 6.8: Pre-committed decision table mapping outcome patterns to dissertation conclusions.

#	Outcome pattern	Dissertation conclusion
2	H1 supported, H2b **fails** (drop > 10 pp)	The system works **for focus, at a quantified situational-awareness cost**, a conclusive, qualified result with a design implication (mode choice by context).
3	H1 **refuted** (90% CI on the paired error-rate difference excludes the SESOI from below)	**The system does not deliver a meaningful benefit** in its core use case, a conclusive negative. The dissertation reports the bounded estimate ("the vignette reduces the error rate by at most X%").
4	H1 estimate positive but CI spans both SESOI and smaller values	The benefit is real-signed but **cannot be claimed at the pre-set size**. Report the estimate with its CI as a bounded conclusion. (Made unlikely by the pilot gate and trial counts. If it occurs, it is reported as such, not spun.)
5	Pilot gates fail after tuning	**The main study does not launch**. The paradigm is redesigned with the supervisor. Inconclusive main-study data is prevented, not explained.

Interpretation of whichever row occurs, including the transfer limits of a positive result and the design implications of a negative one, is developed in Chapter 7. The oracle-vs-live perception gap that bounds the generality of Block B's simulated context is quantified in Chapter 5.

6.10 Operational Protocol and Threats to Validity

6.10.1 Operational protocol

Three properties of the running protocol bear on whether the data can be trusted.

The session is deterministic. Every distractor burst, marker onset, and task trial derives from versioned schedule files in the repository, and condition order derives from the participant ID, so a session is reproducible from those two facts alone. No experimenter arithmetic happens mid-session and no live machine learning runs anywhere. The sequencer asserts the required scene configuration at start and refuses to run on mismatch, which matters because two settings that Block B depends on are off in the stock scene: motion suppression, and baked-detection playback, without which SignPop's colour pop, window, and saliency boost do nothing at all.

Logging is redundant. Telemetry writes one CSV row per frame at roughly 60 Hz, flushed every frame, so an unbroken timestamp chain is itself evidence of integrity and a crash costs under a second of data. Every event is a labelled row. A `scrcpy` screen recording gives a second channel against which any disputed trial can be re-checked.

Data are validated before the participant leaves. A validator checks timestamp continuity, condition counts and durations, and per-condition event counts, for example exactly 40 scored markers per Block B condition, returning PASS/FAIL while the participant is still in the room, so a voided condition can be re-run under the pre-registered rule rather than discovered a week later. Every deviation goes to an incident log with the rule that applied, and those are reported rather than hidden.

6.10.2 Threats to validity

Table 6.9: Threats to validity and their mitigations.

Threat	**Mitigation**
Passthrough acuity confound (Quest 3 VST below normal acuity, worse in low light)	No fine-text reading through passthrough anywhere; the task is rendered on a virtual panel rather than read through passthrough; legibility confirmed at G1; bright constant lighting on the checklist
Novelty effect	Practice-to-criterion tutorial including the vignette before any measured trial; counterbalancing distributes residual novelty
Order and learning effects	Parity counterbalance in both Block A and Block B; matched sheets/segments; equal trial counts. Fixed A→B block order is a limitation: B is always second, acceptable because B's comparison is internal to the block and counterbalanced within it

Table 6.9: Threats to validity and their mitigations.

Threat	**Mitigation**
Coverage limits from the session budget	The three-mode sampler was cut, so Soft Dark was never shown to a participant and no participant saw two modes on a common stimulus (Section 6.7); the mode $\times$ scenario confound is therefore unmitigated, and subjective data is one end-of-session opinion per participant rather than a per-mode comparison
No workload measure	A per-condition NASA-TLX was specified but never administered, so the study has no subjective workload data and cannot say whether the vignette made the task feel easier as well as produce fewer errors. The end-of-session questionnaire asks about perceived focus benefit, which is adjacent but not the same construct, and it is answered once for the whole session rather than per condition
Demand characteristics	Objective primary DVs (errors, millisecond RT, telemetry); neutral "comparing display modes" framing; the noticing question follows Block A data collection
Distractor floor (distractors fail to distract)	Relies on the established distractor-cost literature cited in Section 6.1.1 (Ai et al., 2025, $N = 66$) rather than an internal manipulation check; not independently verified within this study
Simulator sickness and dropout	Seated, no locomotion, $\sim$35 min headset time, mid-session break, VRSQ pre/post, stop rule, $N + 4$ recruitment
Small-N insensitivity	Within-subjects design, high trial counts, mixed models; single primary endpoint; SESOI and equivalence margins make null results informative; effect sizes and CIs always reported
Experimenter variability	One experimenter; verbatim scripts; printed per-session checklist

Table 6.9: Threats to validity and their mitigations.

Threat	**Mitigation**
Ecological validity (driving is a video; workstation distractors are tablets)	The claim tested concerns attention mechanisms under controlled distraction, with driving footage as stimulus context, explicitly not a road-safety claim (Chapter 1 scoping rules); developed further in Chapter 7
Stereo comfort of camera-fed effects	The dark modes are overlay-only (no camera repaint); SignPop runs in the video scene (no live camera fusion, and its detection gate is driven by the offline oracle track, never a live model); head motion is moderate by task design
SignPop's manipulation is a bundle	A confirmed detection switches on six effects at once: colour pop, window carve-out, saturation lift, brightness lift, contrast expansion, and darkened surround (Chapter 4, Section 4.5.5). So H2a and H2b cannot attribute any measured effect to one component. This is not mitigated in the current design. Chapter 8 proposes a three-arm follow-up (no filter, filter with detection off, filter with detection on) to separate them

6.11 Interim Results (N = 14 Block A, N = 13 Block B)

Recruitment continues toward the N = 20 target of Section 6.1.3. Fifteen people have been tested across an initial pilot day and four further collection days. Block A yields **fourteen** usable paired participants and Block B **thirteen**, the two sets overlapping but not identical because some participants contributed a usable result to one block and not the other. What follows is the interim analysis of those pairs. It is reported as interim for three reasons: it is not the pre-registered confirmatory analysis of Section 6.9, achieved power sits near 60% rather than 80%, and recruitment is still open.

Two provenance facts belong with the numbers rather than in a footnote. First, Block A is pooled across the four task versions used during the study, following the analysis decision of 2026-08-06. The versions differ, and a Kruskal-Wallis test across them is significant ($H = 13.59$, $p = .0035$), but the difference is the deliberate v4 change that added lures and switched the distractor reel in response to a 95–100% ceiling on v3. Within the test versions the only timing difference is a 1.8 s against a 2.0 s stimulus onset asynchrony, and median correct reaction time across every run sits between 630 and 950 ms, so no participant is response-limited under either and a 200 ms difference cannot plausibly drive an accuracy gap. Second, and less comfortably, **one of the 28 Block A values is experimenter-reported rather than measured**: P4's no-filter accuracy (98.4%) has no exported CSV. It is retained because the assignment is documented, but it is the only such value in the analysis and every figure below inherits that qualification.

Three participants are excluded from Block B under documented rules, never inferred ones: P3 for 539 false alarms on the filter run with provably looser hits (Mann-Whitney $p = .0301$), P9 by experimenter decision on the day, and P15 for 206 false alarms across 246 presses. Block A is retained for P3 and P9, since the exclusions concern the marker task alone.

6.11.1 H1: Block A, workstation focus

Accuracy on the memory task rose from 92.19% (SD 7.13) with the vignette off to 95.53% (SD 4.58) with Hard Dark on. That is a gain of **+3.34 percentage points** (SD 5.21, median +2.28; 95% CI +0.33 to +6.34, paired $t(13) = 2.40$, $p = .032$; Wilcoxon $W = 19.0$, $p = .035$; $d_z = 0.641$). Ten of fourteen participants improved. The interval excludes zero on both tests, which is the strongest statement this sample supports.

Achieved power at this N and effect size is 60%, and roughly 22 participants would be needed for the conventional 80%. **H1 is supported at the interim** N, with the caveat that the lower confidence bound sits at +0.33 points, close enough to zero that the effect's true size remains unpinned.

6.11.2 H2b: Block B, pooled safety check

Hit rate pooled across the whole field of view rose by **+5.58 percentage points** with SignPop on, from 50.00% (SD 17.02) to 55.58% (SD 9.02), with seven of thirteen participants improving. The paired difference test does not reach significance ($t(12) = 1.51$, $p = .156$; Wilcoxon $W = 21.0$, $p = .175$; $d_z = 0.420$), which is beside the point, because H2b was never a difference test.

H2b's refuting observation did not occur, so the safety claim stands. Section 6.2 committed in advance to what would refute H2b: equivalence not established *and* a point estimate showing a drop greater than 10 points. The estimate is a rise of 5.58 points, so the second condition is not met and the mode is not shown to harm situational

Slopegraph: one line per participant, NoFilter accuracy on the left axis, Filter accuracy on the right, 14 lines, the 10 improvers rising and the 4 decliners falling. Group means overlaid as a heavier line. P4 distinguished by marker or dashed line, since its NoFilter value is experimenter-reported rather than measured.

Figure 6.2: Block A per-participant accuracy, vignette off versus Hard Dark on (N = 14).

awareness. The 90% confidence interval runs -0.98 to $+12.14$, which bounds the worst case the data support at about a one-point loss, far inside the ten-point margin. Peripheral awareness was not degraded.

The pre-registered statistic is nonetheless worth reporting exactly. The two one-sided tests return $p = .0006$ on the lower bound and $p = .126$ on the upper, so formal equivalence within ± 10 points is not established at this N. The reason is entirely one-directional: the data cannot exclude a *benefit* larger than ten points. That is a specification issue rather than a result. H2b asks a one-sided question, whether re-grading costs peripheral awareness, and a two-sided equivalence procedure additionally requires ruling out a large gain, which the safety framing never cared about. The instrument matched to the question is a non-inferiority test against a -10-point margin, which is the lower one-sided test above, and it passes decisively. Both numbers are given here so the reader can see the pre-registered procedure and the question it was meant to serve, and Chapter 7 returns to the lesson. What the dissertation does not do is quietly substitute the non-inferiority reading for the pre-registered one after seeing the data.

6.11.3 H2a: Block B, the 20–30° band

Hit rate broken out by eccentricity band shows a pattern rather than a uniform shift. Every band moves in the same direction, but only one moves reliably. Below 10° the change is $+5.36$ points ($p = .753$), where the colour-pop gate is near zero strength and has little to lift. From 10–20° it is $+4.61$ points ($p = .403$), with the gate at roughly 20% strength. **From 20–30° the change is +16.53 points** (38.6% to 55.1%, t $p = .034$, Wilcoxon $p = .049$, $d_z \approx 0.66$, achieved power 59%). Beyond 30° it is $+1.92$ points ($p = .692$), where the gate is at full strength but the target sits far enough into the

Equivalence plot: horizontal axis is the paired difference in pooled hit rate in percentage points. Shaded band marks the pre-registered ±10-point margin, with the −10 safety bound emphasised. Point estimate at +5.58 with the 90% interval (−0.98 to +12.14) drawn as a horizontal bar, sitting clear of the safety bound on the loss side.

Figure 6.3: Block B pooled hit rate against the safety margin (H2b, N = 13).

periphery that a stronger signal buys almost nothing.

The 20–30° band is the only one individually reliable on either test, and it is between three and eight times larger than any other band. **H2a is supported at the interim N, specifically and only in the band the mechanism predicts**, as a secondary, exploratory analysis rather than a substitute for a confirmatory test at the final N.

Two limits belong beside that finding. Band membership uses each target's mid-lifetime position, which is a weak proxy: targets move, the median lifetime swing is 17.2°, and roughly 31% of catches land in a different band from the one the mid position assigns. And with all four bands now pointing the same way, the honest reading is a general positive trend with one band carrying almost all of the signal, not three null bands and one live one.

Grouped bar chart, the dissertation's headline result. Four eccentricity bands on the horizontal axis (<10°, 10–20°, 20–30°, >30°), paired NoFilter and Filter hit-rate bars per band with 90% CI whiskers. The 20–30° pair carries the +16.5-point gap and is the only band whose interval excludes zero. A second panel or overlaid line shows colour-pop gate strength across the same axis, with the useful-field-of-view boundary at roughly 30° marked, so the alignment between where the gate acts, where the eye can still use the result, and where the benefit appears is visible in one figure.

Figure 6.4: Block B hit rate by eccentricity band, with gate strength and the useful-field boundary overlaid (H2a, N = 13).

6.11.4 What this interim result does and does not show

The manipulation-is-a-bundle limitation (Section 6.10.2) applies in full. Nothing above attributes the 20–30° gain to the colour lift, the detection aperture, the saturation boost, or the darkened surround individually.

Thirteen end-of-session questionnaires (Section 6.7) show a divergence worth flagging rather than resolving here. Perceived focus benefit is strongly positive, 57 of 65 responses favouring the effects. Perceived awareness cost splits close to evenly at 38 of 65, so roughly four in ten responses report feeling cut off from the surroundings or noticing side events late. Visual comfort sits at 38 of 52 across its four answered items. The three items on holding attention, looking away, and knowing where to look were answered favourably by 12, 11 and 12 of the thirteen respondents. At $N = 10$ these three were unanimous and they are no longer, which is a useful reminder of how little weight a unanimous result at that sample size could carry.

The subjective impression of lost awareness is not supported by the behavioural hit-rate data, which shows no loss at all. Chapter 7 takes up what to make of that gap.

Diverging stacked bar, one row per end-of-session construct: perceived focus benefit (57/65 favourable), perceived awareness cost (38/65), visual comfort (38/52 across four answered items). Favourable responses extend right of centre, unfavourable left, so the split shape of the awareness construct contrasts against the near-solid focus-benefit bar.

Figure 6.5: End-of-session questionnaire responses by construct (N = 13).

Achieved power throughout this section is 60% for H1 and 59% for the 20–30° band, so a null result at this N would not have been informative either way. The results reported here cleared their thresholds regardless, which is not the same claim as having been powered to find them.

Chapter 7

Discussion

Status note. This chapter was originally drafted before the user study (Chapter 6) had produced any data. Most of its interpretive frame (Sections 7.3–7.4) is kept in that pre-committed form deliberately. It is the discussion-chapter counterpart of the pre-registered decision table, fixed before data collection so results cannot be interpreted by a frame built afterwards to flatter them. The technical benchmark programme of Chapter 5 still has not produced data beyond the offline detection bake. The user study, however, now has interim results from fourteen Block A pairs and thirteen Block B pairs (Chapter 6, Section 6.11). Section 7.2 below discusses them against the pre-committed frame that follows it. This is an interim discussion, not the confirmatory one the frame was built for, and it is named as such throughout.

7.1 What can already be discussed

7.1.1 The design space is itself a result

The central finding of Part T does not await any benchmark: it is the shape of the design space in Chapter 3. On a consumer VST headset, diminished reality is not one problem but three, indexed by compositing access. At Tier 1, the platform grants global colour remapping of the system passthrough layer (the Styling API's brightness, contrast, saturation, and LUT controls) and nothing spatial: no per-pixel masks, no world-locked regions, no read access, and, since the deprecation of surface-projected passthrough at SDK v83, no per-surface control either. The significant fact is the *absence*: a developer who wants to dim the periphery and spare a window cannot do it at Tier 1 at all. That absence is a platform-policy finding, not an engineering one, and it shapes what DR research is possible on consumer hardware more than any algorithm does.

The system's response to that absence, the world-locked focus window realised as a shaped *absence* of overlay, native passthrough showing through it, with all manipulation confined to the periphery, is an existence proof: windowed subtractive DR is achievable at Tier 2 without OS cooperation, and hybrid Tier-2/Tier-3 composites (native passthrough inside the window, an effect-processed camera feed outside it) are achievable with Passthrough Camera Access. The structured prior-art search reported in Chapter 2, Section 2.5.2 found no publication, product, or open-source project that composites system passthrough with a live shader-processed camera feed in one view, for any purpose, while every constituent mechanic has documented precedent. The composite, precisely scoped, is the novelty claim, and the documented search protocol is what entitles the dissertation to make it. Should the refresh search before submission surface a concurrent system, the claim degrades gracefully to independent invention with a characterised design space, which is why the claim is worded around the space rather than the trick.

7.1.2 Engineering lessons that generalise

Four lessons from the implementation are candidates for reuse beyond this project.

Sample by world direction, not screen space, when re-rendering a mono camera feed. Screen-space UV mapping of the camera texture gives each eye a different pixel for the same world point, and the result is double vision. Sampling by world direction through a head-centred pinhole is eye-independent, so the binocular system fuses the re-rendered image. Any future system that re-renders a mono VST feed, on this platform or another, will face the same problem and can use the same solution.

Exploit the quality asymmetry: route the task-critical region through the native layer. The naive architecture re-renders everything and accepts camera-feed quality everywhere. Inverting it, manipulating only the periphery and leaving the window as an absence, means the user's task-relevant vision never pays the Tier-3 quality tax. Chapter 5's T3 benchmark will quantify the tax. The architectural lesson stands independently: on any platform where an accessible feed is worse than the native view, put the manipulation where the eyes are not.

Decouple effects from head motion. Head-coupled peripheral restriction increases discomfort (Norouzi et al., 2018). The system therefore fades the effect out during fast head rotation and restores it on settling, with an immediate restore when the head enters the focus region. This inverts a common default, effects that follow the head, into one that yields to the head.

Consumer-platform DR research inherits platform-policy risk. The failure museum's least technical entry may matter most: surface-projected passthrough, a capability this project might have built on, was deprecated mid-project at SDK v83. Tier boundaries on consumer hardware are set by vendor policy and move without notice. Research designs that, like this one, keep a working path at every tier are robust to that movement. Designs premised on a single capability are not.

7.1.3 Positioning: software-defined visual noise cancellation

The literature review's closest conceptual frame for the whole project is visual noise cancellation (VNC): moderating the visual environment through head-worn hardware the way acoustic noise cancellation moderates sound. Read through that frame, the present system is a *software-defined* VNC device: the "cancellation profile" is a shader parameter set, the modes are profiles (attenuate: Soft/Hard Dark; re-grade: SignPop), and the tier taxonomy describes which profiles the platform physically permits. The analogy also imports a useful discipline from its acoustic original: noise-cancelling headphones are evaluated on measured attenuation *and* on what they cost the wearer in awareness, which is exactly the H1/H2b pairing. The analogy is offered as positioning, not as a claim of equivalence: acoustic cancellation is subtractive at the signal level, whereas everything in Chapter 3 exists precisely because consumer VST compositing forbids signal-level subtraction and forces synthesis by overlay.

7.1.4 What could not be discussed before, and what changed

Before any data existed, no statement about whether the system *works*, in the sense of H1, H2a, or H2b, was available. The literature gave reason to take both outcomes seriously. Peripheral distractors reliably impose sustained-attention costs in headsets (Ai et al., 2025), and DR-style attenuation reduced workload in VR simulation (McLaughlin et al., 2025). But area darkening's guidance effects come mostly from immersive video contexts, desaturation alone has shown weak effects (Wang et al., 2020), and awareness costs are documented wherever attenuation succeeds (Murph et al., 2021). The design space contained both a working system and a placebo, and only data could say which. Fourteen Block A pairs and thirteen Block B pairs now exist (Chapter 6, Section 6.11). The next section discusses what they show, against the interpretive frame that follows it, as an interim reading rather than the confirmatory one that frame was built for.

7.2 Interim results, discussed

The interim data point toward the working-system end of that range, on both questions the design set out to ask, with a genuine surprise attached to the second.

The workstation benefit (H1) is present and sizeable at this N. There is a 3.34-point accuracy gain (95% CI +0.33 to +6.34, $p = .032$, $d_z = 0.641$) on a real-hardware task, with real physical distractors. It survives both a parametric and an assumption-free test, and ten of fourteen people improved individually. The honest caveat is the interval's width. A lower bound of +0.33 points is compatible with a benefit barely worth the SESOI the pre-registered design set at 50% of the vignette-off error rate. At 60% power the true effect could plausibly be smaller or larger than +3.34 once the remaining participants are collected. It is worth noting that adding three participants moved the estimate by only a third of a point while tightening nothing much, which is what a real but modest effect looks like as a sample grows.

The safety question (H2b) comes back clean. Pooled hit rate did not drop. It rose by 5.58 points, and the 90% interval (−0.98 to +12.14) puts the worst case the data support at about a one-point loss, far inside the ten-point margin the design set. Section 6.2 committed in advance to what would refute H2b, a drop greater than ten points alongside a failure to establish equivalence, and the estimate points the other way, so the refuting observation did not occur. Nothing here suggests that re-grading salience costs the user their periphery.

One methodological lesson comes out of the statistic rather than the result. Formal equivalence within ±10 points is not established at this N ($p = .126$), and the reason is instructive. The two-sided procedure requires ruling out a large *benefit* as well as a large loss, and the data cannot rule out a benefit above ten points. But a large benefit was never a safety concern. H2b asks a one-directional question, so the instrument matched to it is a non-inferiority test against a −10-point bound, which is the lower one-sided test and passes at $p = .0006$. The mismatch is a specification error made when the hypothesis was written, not a property of the system, and it is worth recording because equivalence testing is increasingly recommended for exactly this kind of safety claim (Lakens, 2017) and is easy to reach for in its two-sided default form when the question at hand only has one side. A study writing a safety bound should state the direction it cares about and test that direction alone.

The genuine surprise is where H2a's benefit lives. The pre-registered hypothesis asked whether SignPop speeds detection generally. What the interim data show instead is a benefit concentrated specifically in the 20–30° eccentricity band, the one region where the colour-pop gate has visible room to act and the target is still close enough to use the result (+16.5 points, $p = .034$). Every band moves positively, but the other three move by between 1.9 and 5.4 points and none is individually reliable, so one band carries almost all of the signal. This was not the shape anticipated when the hypothesis was pre-registered. It emerged from breaking the pooled result down by angle, which makes it a secondary, exploratory finding rather than the confirmatory test H2a was designed to be.

What makes it more than a lucky slice is that the band was fixed by the design before any data existed. The soft edge was set at 20° against the useful field of view, the roughly 30° radius within which people extract task-relevant information without moving their eyes (Ball & Owsley, 1993; Chapter 4, Section 4.2). The benefit therefore appears in precisely the annulus between where the gate starts having room to act and where the useful field runs out. Inside 20° the gate has nothing to reveal, and beyond 30° it reveals plenty to an eye that can no longer use it. This is exploratory analysis agreeing with the system's own geometry rather than fitting noise, which is the strongest kind of exploratory finding available at this sample size, and it is named as exploratory every time it is used in the rest of this chapter.

The subjective and behavioural data partly disagree, and the disagreement is itself informative. Perceived focus benefit is strongly positive, 57 of 65 responses. But roughly four in ten responses on the awareness construct report feeling cut off from the surroundings or noticing side events late, against a behavioural hit rate that rules out any large peripheral-awareness loss and, in the 20–30° band, shows an improvement. Two explanations are worth taking seriously rather than explaining away. First, noticing that the periphery has been changed is a different experience from missing something in it, and every DR manipulation studied to date is noticed (Sutton et al., 2022). "I can tell my vision has been altered" and "I failed to detect something" are not the same report, even when only the first is true. Second, narrowed-field anxiety is documented independently of task performance. It rises in every restricted-field condition tested in the literature, including conditions where spatial learning is unaffected (Barhorst-Cates et al., 2016). Reassuring people directly, for instance with a visible cue when something happens outside the window, would reintroduce exactly the salience the filter exists to remove. That is the trade-off in its sharpest form, and it is left unresolved here.

7.3 The pre-registered interpretation frame

The decision table (Chapter 6, Section 6.9.3) is reproduced here in condensed form, with the scientific meaning each row would carry. The dissertation will report the row that occurred once the final $N = 20$ collection concludes. This section exists so that the meaning of each row is on record before that data is. The interim data discussed in Section 7.2 sit closest to row 1, but Section 6.9.3 is explicit that this table maps the confirmatory collection, not an interim reading of it.

Row 1, H1 supported and H2b passes equivalence, would validate the mechanism claim on real hardware. The claim would remain mode-in-context, Hard Dark at a workstation and SignPop on a monitoring stimulus, and the design cannot support "dark vignettes work" in general. Within that scope it would be the first performance-based demonstration of DR distraction suppression on consumer passthrough hardware rather than in VR simulation.

Row 2, H1 supported and H2b failing, is not a failure row. It would reproduce and *quantify* on real hardware the focus–awareness trade-off documented in simulated DR (McLaughlin et al., 2025; Murph et al., 2021), which is arguably the most design-relevant outcome available, because it converts mode choice from an aesthetic decision into a safety-relevant one: attenuation modes for closed environments where the periphery genuinely does not matter, salience modes or nothing where it does.

Row 3, H1 refuted, would be the bounded negative, and it carries a specific scientific reading. If peripheral visual attenuation at Hard Dark's maximum reduces the error rate by less than the SESOI, then distractor interference in this paradigm is not carried solely by continuing visual input. Knowing the tablets are there, and holding an attentional set toward them, may impose costs no visual overlay can remove. That would be genuinely useful to a field that currently extrapolates from simulation studies in which removing a distractor is total.

Rows 4 and 5 exist to forbid two temptations: retrospectively deciding that a smaller effect "would also be meaningful", and omitting an abort path from a pre-registered frame.

7.4 Block A versus Block B: a built-in test of simulation-based evaluation

The two confirmatory blocks run the same guidance principle at opposite ends of the feasibility spectrum: Block A on the real Tier-2 artifact over live passthrough with physical distractors, Block B on the simulated context, video sphere, scripted probes, that stands in for the future system. This pairing was designed as a bridge, and it carries an incidental methodological payload. If the two blocks agree in direction, diminishment helps focal performance in both, the simulation methodology that the nearest published work relies on entirely (Cheng et al., 2022; McLaughlin et al., 2025, both VR-simulated) gains a point of external credibility from this project's real-hardware anchor. If they diverge, the divergence localises what simulation misses: candidate explanations would include the absence of depth motion in monoscopic video, the different distractor ontology (baked footage events versus physically present screens), and hardware-borne effects such as fusion strain that no video condition reproduces. Either way the comparison is reportable, which is the property that was designed for. The blocks were not powered as a formal crosscontext contrast, and any A-versus-B statement will be framed as observational.

Operationally, "agreement" will be read at the level of sign and decision-table row, not effect magnitude: Block A's H1 and Block B's H2a estimate the benefit of diminishment-family manipulations on focal performance in their respective contexts, and the comparison asks whether both point the same way. Magnitudes are not comparable across the blocks (different tasks, different dependent variables, different modes), and no ratio between them will be interpreted.

7.5 Transfer: what leaves this dissertation, and under what conditions

To optical see-through (OST) hardware, the dark overlay does not transfer. The Tier-2 modes work because a VST display is an opaque screen under full render control. Additive OST optics cannot darken the world. What transfers is the taxonomy and the question. On OST hardware the taxonomy collapses toward display-side attenuation, of which segmented dimming hardware (e.g. Magic Leap 2's segmented dimmer (Magic Leap, 2024), a dedicated low-resolution panel that locally attenuates light behind masked virtual content) is the nearest existing analogue of a Tier-2 windowed dim, and toward the additive salience modulation demonstrated by Sutton et al. (2022). Whatever Chapter 6 finds about *what peripheral diminishment does to attention* transfers as a human-factors result to any hardware that can implement an equivalent stimulus. The implementation findings transfer only within VST.

From the simulation, transfer is conditional, and the conditions are enumerated. Headset weight biases only comfort ratings, and conservatively. Attention effects are unaffected in either direction by lightening hardware. Monoscopic video removes motion-in-depth, so peripheral capture by looming stimuli is untested. The passenger viewpoint removes task coupling, which is why no statement in this dissertation concerns driving performance, and none should be quoted as if it did. Video dynamic range compresses glare, so SignPop's glare-suppression track is exercised only weakly. Identical stimuli for all participants trade ecological breadth for internal validity, and the trade is intended. Block A, running on the real system against physical distractors, is the one component that transfers without these caveats, which is why it carries the primary hypothesis.

Oracle conditioning. Every statement about detector-dependent guidance variants is conditioned on the oracle assumption, whose measured size is Chapter 5's T4 deliverable. Until the live column of the oracle-gap table is filled, "the detector sees everything in time" is an assumption with a plan for its own measurement, not a fact. The study protocol is not insulated from this caveat: SignPop, Block B's evaluated mode, is itself detector-dependent, on offline, oracle-quality detection rather than a live pipeline, so RQ2/H2a/H2b inherit the same conditioning as any future detector-dependent extension, not a lesser version of it.

7.6 Limitations, consolidated

The limitations below are collected from Chapters 3–6 so that they appear once, together, rather than diluted across the text.

1. **Mode × scenario confound.** Hard Dark is tested only in the workstation context, SignPop only in the monitoring context. Every empirical claim is mode-in-context. The design cannot separate mode effects from scenario effects, and nothing in the design mitigates it (Chapter 6, Section 6.7).

2. **SignPop's manipulation is a bundle.** A confirmed detection switches on six effects at once: colour pop, window carve-out, saturation lift, brightness lift, contrast expansion, and darkened surround (Chapter 4, Section 4.5.5). None is isolable in the current design, so RQ2/H2a/H2b cannot attribute a measured benefit to any one of them.

3. **Soft Dark carries no participant data at all.** It was never placed in front of a participant (Chapter 6, Section 6.7). Its place in the dissertation rests on design-space grounds (Chapter 3, Section 3.5) and on its shader path, never on evidence about how it is experienced.

4. **No eye tracking.** Quest 3 has none. Head pose is a validated proxy at the eccentricities used (Higgins et al., 2022; Sitzmann et al., 2018), but covert attention shifts, distraction without head movement, are invisible to telemetry. The dependent variables were chosen so that *performance outcomes*, not attention allocation, carry the hypotheses. The mechanism remains partially inferred.

5. **Sensitivity floor.** N = 20 gives 80 % power at dz ≈ 0.65 for the classical paired test. Trial-level mixed models improve on this, but effects meaningfully smaller than the SESOI are detectable only as estimates with wide intervals (decision-table row 4 exists for this reason). At the interim sample, achieved power is 60% for H1 ($N = 14$) and 59% for the 20–30° band ($N = 13$), below the pre-registered target for

either. That is why both are reported as interim rather than confirmatory findings, regardless of clearing their significance thresholds.

6. **Single experimenter, single site, single hardware generation.** Scripts and checklists bound experimenter variance but cannot eliminate it. Every number is specific to Quest 3 and the frozen OS/SDK versions, on a platform whose capabilities have already moved once mid-project.

7. **Ecological validity boundaries.** The workstation distractors are two tablets with scheduled bursts. The monitoring stimulus is a video. The study measures attention mechanisms under controlled distraction, deliberately, and its conclusions end where those controls end.

8. **Residual novelty.** Practice-to-criterion and counterbalancing distribute first-exposure effects. They do not abolish the fact that every participant meets peripheral diminishment for the first time that hour. Longitudinal adaptation, the deployment-relevant question, is future work by construction.

9. **Passthrough acuity boundary.** The task was designed around the platform's acuity ceiling (virtual panel or pilot-verified large print), so no claim extends to fine-detail real-world tasks such as reading small text through passthrough, the very context that motivated some of the use cases. This is a hardware boundary the dissertation documents rather than solves.

7.7 Implications in both futures

If the system works (rows 1–2). The immediate implication is that subtractive attention guidance is deliverable *today*, at Tier 2, on unmodified consumer hardware: no OS cooperation, no eye tracker, no detector. The near-term research programme that opens is concrete: hour-scale field deployment of the camera-free dark modes (whose endurance T5 tests), the temporal-transition and user-controlled-release questions already identified as gaps in the literature review (gaps 3 and 4), an eye-tracked replication to convert performance findings into attention-allocation findings, and an OST translation via segmented dimming hardware. A row-2 outcome adds the design rule that mode selection is a safety decision, and would motivate a context-detection layer, which is where the oracle-gap measurement re-enters as an engineering budget.

If the system does not work (row 3). The negative is bounded and useful. For the field, it would indicate that peripheral visual attenuation on passthrough, the cheapest and most deployable DR, does not by itself buy back distraction costs, redirecting effort toward the channels the overlay cannot touch: semantic knowledge of the distractor, auditory leakage, and interruption timing. For this dissertation, the contributions that survive are exactly the ones constructed to be result-independent: the design space and its tier constraints (C1), the reference implementation and its transferable engineering lessons (C2), the benchmark characterisation of a consumer VST platform (C3), and a pre-registered, falsifiable evaluation method for DR attention claims (C4), which a null result would demonstrate rather than undermine. A field that currently argues DR's promise largely from simulation would gain its first hard, hardware-grounded boundary.

In either future, the method is part of the contribution. The evaluation apparatus built for this dissertation, a single primary endpoint with a smallest effect size of interest, equivalence margins for safety claims, a gated pilot whose failure aborts rather than degrades the main study, and a decision table committed before data collection, is not standard practice at masters-project scale, where underpowered multi-hypothesis designs that can only end inconclusively remain common. The apparatus costs little (its artefacts are documents and thresholds, not equipment) and converts every outcome into a claim with stated bounds. Whichever row of the decision table occurs, the dissertation demonstrates the apparatus by using it in public, which is a transferable contribution to how small-N XR evaluations can be run.

Both futures are written here with equal seriousness because the study was designed so that either one is a finding. That property, not any particular row of the decision table, is the standard this dissertation set out to meet.

Chapter 8

Conclusion and Future Work

8.1 The question, revisited

This dissertation asked one question with two halves: *can subtractive attention guidance be delivered on consumer XR hardware, and when it is delivered, does it actually help human attention?*

The first half has been answered in the affirmative, with precision about what "delivered" means. On the Meta Quest 3 the question reduces to a compositing-rights problem, and the access an application has stratifies into three tiers: colour-only styling of the OS-owned passthrough layer, overlay compositing on top of it, and full pixel control over a lower-quality camera feed (Chapter 3). That narrow space turns out to be sufficient for a working multi-mode system, built around an architectural inversion in which the focus region is rendered as an *absence* in an overlay, so native passthrough at full quality serves the region that matters while the treatments occupy the periphery around it (Chapter 4). The hybrid composite at the heart of the most capable mode survived a structured prior-art search with no published or shipped precedent found (Chapter 2).

The second half has an interim answer and a confirmatory one still pending. The pre-registered design carries one properly powered primary hypothesis, that peripheral dimming reduces the cost distractors impose on a focal task, and an equivalence-bounded safety hypothesis, that salience re-grading must not degrade peripheral awareness beyond a stated margin. Fourteen Block A pairs and thirteen Block B pairs exist today, and they show a 3.34-point workstation accuracy gain, peripheral awareness that was not lost, and a detection benefit concentrated exactly in the eccentricity band the system's own detection-gated design predicts. Every part of that is qualified by the sample it comes from, at roughly 60% achieved power, well short of the pre-registered target. But "inconclusive" is still not among the reachable outcomes. The interim data point in a specific, mechanistically coherent direction, and the remaining collection will either confirm that direction with tighter intervals or correct it.

8.2 Contributions, honestly stated

Four contributions were claimed in Chapter 1; their standing at the time of writing is as follows.

C1: the design space (Chapter 3) is complete as an analytical contribution and is grounded in implementation experience rather than speculation. Its quantitative columns (frame cost, latency, legibility per tier) await the Chapter 5 benchmark campaign.

C2: the reference implementation (Chapter 4) is complete and running on target hardware: the world-locked focus-window architecture, world-direction fusion sampling for the monocular feed, motion-coupled suppression, and ten peripheral treatments spanning the design space, of which two, SignPop and Hard Dark, are carried to confirmatory evaluation, together with an engineering record that reports dead ends, screen-space sampling double vision, a deprecated surface-projection API, shader stripping on Android, as reusable knowledge.

C3: the technical evaluation exists as a specified plan with preliminary results (the offline full-resolution detection bake: 13,939 detections across 1,040 sampled frames, 99.5% of samples with at least one detection). The benchmark campaign itself, frame timing, latency, legibility, thermal endurance, and the oracle-versus-live gap, remains to be run.

C4: the pre-registered study design is complete, ethics-mapped against the approved protocol, and instrumented on paper down to verbatim participant scripts and exclusion rules. Data collection is underway. Eleven of the target twenty participant pairs are in, with interim results reported in Chapter 6, Section 6.11 and discussed in Chapter 7, Section 7.2. Recruitment continues toward the full sample, and the confirmatory analysis and decision table of Chapter 6, Section 6.9.3 apply once it is reached.

An examiner should read C3 and C4 as what they are: the empirical programme of this dissertation, in a state where every analytical decision has been made and committed to in advance of the data. That is precisely the state in which such decisions carry evidential weight.

8.3 Future work

Seven directions follow directly from the material.

Separating the bundle. The interim 20–30° finding (Chapter 6, Section 6.11) is the clearest reason this item leads the list. A confirmed detection switches on six effects at once: colour pop, window carve-out, saturation lift, brightness lift, contrast expansion, and darkened surround. So nothing in this dissertation attributes the +16.5-point gain to any one of them. A three-arm design, no filter, filter with the detection gate off, filter with it on, would need no new code, since the gate already has a toggle. It would directly separate "dimming helped" from "passing the signal through helped."

Closing the oracle gap. SignPop, one of the two evaluated modes, is detector-dependent today, on offline, oracle-quality detection, not a live pipeline, and Chapter 5's benchmark quantifies how far live on-device detection currently falls short of that oracle. The engineering programme this implies, quantised detectors, region-of-interest second passes, temporally amortised inference, is well charted. The natural next system iteration replaces SignPop's offline oracle gate with a live detector and re-runs Block B to test whether the measured benefit survives the live-detection accuracy and coverage gap this dissertation deliberately left unclosed.

Transfer to optical see-through hardware. The Tier-2 dark overlay does not transfer to OST glasses, whose additive optics cannot darken the world. This was stated in Chapter 7 and bears repeating as a boundary, not a defect. What does transfer is the taxonomy itself, on OST hardware the design space collapses towards display-side dimming layers and segment attenuation, and, critically, the human-factors findings, which concern peripheral diminishment as a perceptual manipulation rather than any one rendering path. A replication of the Chapter 6 paradigm on a dimming-capable OST device would be the cleanest possible test of that transfer.

Adaptive introduction of diminishment. An idea raised and deliberately deferred during study design: rather than switching the effect on, a deployed system might introduce it gradually as focus wanes. As a validation instrument a ramp confounds time with intensity and was excluded from the study. As an *adaptive-system UX question*, when and how fast should a focus aid fade in, and should the user or the system control it, it is a well-formed research problem, seeded by the system's existing formation animation and motion-coupled fade mechanics, and by the pre-registered time-on-task analysis in the study, which will show whether the vignette's benefit grows as boredom sets in.

Dose–response. The study tests only the extreme of the attenuation axis (Hard Dark). The midpoint, Soft Dark, was built but never tested with participants, so the middle of the axis has no data behind it. A discrete-levels design, off, partial, full attenuation, counterbalanced, would establish the shape of the benefit curve and answer the deployment question this dissertation cannot: how much diminishment buys how much focus, at what comfort cost.

Hour-long deployment. The published thermal ceiling on continuous on-device camera processing (throttling within five to ten minutes) makes the dark modes, which have no camera pipeline, the only plausible basis for session-length deployment today. A longitudinal study of real work sessions under one of them, with the study's workload and comfort instruments administered across days rather than minutes, is the necessary bridge between this dissertation's controlled findings and any claim about practice.

Saliency-weighted diminishment. The literature review identified the combination of channels (desaturation with blur; dimming weighted by computed scene saliency) as unstudied in AR attention guidance. The system's shader architecture already computes per-fragment treatment. Driving its intensity from a saliency model of the periphery, dimming most where the periphery pulls hardest, is a natural composition of the Tier-3 rendering path with the saliency-modulation literature, and a candidate for the strongest version of the guidance concept.

8.4 Closing

The premise of this work was that a consumer headset can be an instrument for *removing* the world as well as adding to it, and that both halves of that proposition deserve rigour: the systems half, where an OS-owned display layer had to be subtracted from by purely additive means, and the human half, where "it helps you focus" had to be converted from a hope into a hypothesis that could fail. The system exists and runs on target hardware. The hypothesis has moved past waiting: eleven participant pairs have already tested it, and the interim direction is favourable, though the confirmatory answer at the full N = 20 sample is still being collected. Whichever way the final data fall, the question will have been asked properly, on the hardware people actually own.

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Appendix A

Study Protocol Artefacts

This appendix holds the operational material of the Chapter 6 protocol: the verbatim scripts read to participants, the pass/fail gates the pilot had to clear before the main study launched, and the pre-registered exclusion rules. It is separated from Chapter 6 so the chapter can argue the design while the artefacts that make the design reproducible remain available in full.

A.1 Participant instructions (verbatim)

> "A panel in front of you will show one shape at a time. **After the first shape, answer whether each new shape is the same as the one before it**, one controller trigger for yes, the other for no. Respond as fast as you can without guessing. Ignore everything else in the room, only the panel matters. Each round lasts about four and a half minutes. There are two rounds. The rounds may look different from each other. Just do the same thing every round."

Participants are never told which conditions belong to "our system" or what the vignette is expected to do. The session framing throughout is that they are "comparing display modes", a defence against demand characteristics (Section 6.10.2).

A.2 Participant instructions (verbatim)

> "You'll watch a driving scene. Small **ring markers will appear briefly** on things in the scene, sometimes in front of you, sometimes off to the side. **Press the trigger as soon as you notice one. That's the only thing you need to do.** Don't press for anything else. Keep watching the road ahead as if you were the driver, and stay seated."

A.3 Pilot study with pass/fail gates

A pilot with n = 3–5 participants (excluded from the main sample) runs in the first week. Its purpose is structural: to make an inconclusive main study impossible. The main study does not launch until all three gates pass. Pilot data never enters the main analysis. Stimulus parameters may be tuned between pilot participants, but hypotheses, margins, and analysis choices may not.

Table A.1: Pilot study pass/fail gates, tests, criteria, and actions on failure.

Gate	**Test**	**Pass criterion**	**On fail**
G1: task form	The forced-choice 1-back tried through passthrough	Panel fully legible; accuracy off both ceiling and floor across pilot participants	Retune timing/difficulty (SOA, answer window, lure ratio) against pilot data, then re-test. Redesign the task only if retuning cannot bring it off ceiling or floor
G2: marker calibration	Block B with candidate marker fade timing and size	Baseline hit rate 70–90% across the sample; unimodal RT distribution	Adjust marker timing/size, then re-test
G3: protocol dry run ×2	Two full end-to-end sessions with the final configuration	Timeline within ±5 minutes; zero log gaps (validator clean); battery ≥ 20% at end; no sickness terminations; questionnaire flow smooth	Fix the specific failure, then re-run one dry run

The pilot concludes with a one-page **lock-in memo** to the supervisor recording the chosen task form, the tuned stimulus parameters, and the finalised absolute SESOI for H1. After this memo, nothing changes until data collection ends.

A.4 Pre-registered exclusion and data-quality rules

Table A.2: Pre-registered exclusion and data-quality rules.

Rule	**Action**
VR-sickness termination	Session ends under the stop rule; partial data retained but participant excluded from confirmatory analysis; replaced from spares
App crash / restart during a condition	That condition void; re-run once at the end of its block if time allows; otherwise the participant contributes remaining conditions to the LMM (which tolerates missingness) and is excluded from the paired robustness test
App crash / restart spanning a **new session launch**, since the ledger assigns a fresh participant ID on relaunch rather than resuming	The two IDs are reconciled into one canonical participant record before any analysis is run. Cleanly completed blocks are joined under the canonical ID, and a block attempted under both keeps only the later complete attempt. The join, both IDs, and the ledger timestamps justifying it are recorded in the incident log and not revisited afterwards
Task accuracy $<$ 60% in NoFilter	Task not performed; exclude and replace
Block B: false-alarm rate $>$ 30% of presses in either condition	Response strategy invalid (indiscriminate pressing); exclude Block B for that participant
Log integrity failure (gaps $>$ 1 s or missing condition markers)	Affected condition void; incident log entry; same re-run rule as crashes
Participant recognises the driving footage	Noted; excluded only on reported strong familiarity affecting behaviour